
ACE: Language-Model Priors and Lookahead Selection for Active Causal Mechanism Estimation

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Abstract

Experimental design is a sequential decision problem: each intervention reveals information that should shape the next. We present Active Causal Experimentalist (ACE), which learns this policy from two ingredients — a pretrained language-model proposer encoding a soft prior over plausible interventions, and a *lookahead* step scoring each candidate against a cloned learner before one is executed — with a preference-based calibration rule (DPO) refining the prior toward the system at hand. Our central finding concerns how little the third ingredient contributes. On a 30-node structural causal model, disabling DPO while keeping proposer and lookahead matches the full method to within seed variance (1.67 ± 0.19 vs. 1.95 ± 0.77), and the two stay level when node names are anonymized to strip semantics. Expanding that comparison from three seeds to eight sharpens it: the five added seeds all exhibit unstable preference optimization and fall back to the passive-baseline plateau, making the calibration step not merely dispensable but a source of variance the rest of the method lacks. Because an intervention budget admits more than one accounting — interventions executed, or every environment query consumed including lookahead probes — we instrument all methods with a query counter and find the asymmetry is not ACE’s alone: Max-Variance issues $65\times$ and Bayesian OED on the order of $500\times$ more hidden oracle queries per decision than ACE’s $4\text{--}5\times$. Rerunning every method at matched *total* query cost, ACE wins by 76%

at 5 nodes and 43–50% at 30, with a zero-hidden-query variant winning while consuming $3.7\text{--}5.2\times$ fewer queries than the baselines it beats. We release the instrumentation, the matched-budget training mode, and the three calibration rules we compare.

1 Introduction

Experimentalists face vast possibility spaces under limited budgets — all pairwise combinations of 100 compounds are 4,950 experiments — and the goal is not prediction but causal understanding — identifying which parameters drive outcomes, separating causal pathways from spurious correlation, and finding intervention targets that generalize. This requires Rung-2 interventions in Pearl’s hierarchy (Pearl, 2009; Pearl and Mackenzie, 2018): controlled $\text{do}(V_i = \nu)$ operations exposing structure not identifiable from observation alone (Spirtes et al., 2000; Peters et al., 2017; Reichenbach, 1956). Theoretical bounds on how many interventions are needed (Eberhardt et al., 2005; Hauser and Bühlmann, 2012) say little about which to run next, and static heuristics — random, round-robin, greedy information maximization (Murphy, 2001) — cannot transfer insight between systems or adapt to what has been learned. Bayesian active causal discovery (Toth et al., 2022; Tigas et al., 2022) adapts, but maintains explicit graph posteriors at substantial cost and targets structure discovery rather than the mechanism estimation we study.

We present **Active Causal Experimentalist (ACE)**, which learns experimental design as a sequential policy: a pretrained language model proposes candidate interventions, a lookahead step scores each by its simulated effect on a cloned learner, and DPO (Rafailov et al., 2023) calibrates the proposer from pairwise comparisons between candidates. This paper’s central move is to *disentangle* these three components empirically rather than present them as an inseparable bundle — and to instrument ACE and every baseline so the environment-query cost of doing so is auditable

rather than asserted.

Contributions.

1. **The LM prior plus lookahead, not DPO, drives the advantage.** On a 30-node SCM, ACE without DPO (same proposer and lookahead, weight updates disabled) matches full ACE to within seed variance (1.67 ± 0.19 vs. 1.95 ± 0.77), and the two stay level under node-name anonymization that strips all semantics (1.49 ± 0.18 vs. 1.51 ± 0.16). Expanding this comparison from three seeds to eight makes the point more sharply: the five added seeds all show unstable preference optimization and fall to the passive-baseline plateau, so the calibration step is not merely dispensable but a source of variance the rest of the method lacks (Section 4.2).
2. **Budget accounting made auditable, then settled.** An environment-query counter on ACE and all five baselines shows ACE’s lookahead cost ($4\text{--}5\times$) is smaller than Max-Variance’s ($65\times$) and Bayesian OED’s ($\sim 500\times$); rerun at matched *total* query cost, ACE wins by 76% at 5 nodes and 43–50% at 30, and the zero-hidden-query variant wins while using $3.7\text{--}5.2\times$ fewer queries (Section 5).
3. **Competitive under conventional accounting.** On the 5-node benchmark ACE reaches median total MSE 0.61 against 1.99 for a Bayesian optimal-experimental-design baseline at matched executed interventions (69%), with the same ordering on coupled Duffing oscillators (0.042 ± 0.036 vs. Random 0.245 ± 0.121).
4. **Emergent strategy.** Without instruction the policy concentrates 99.8% of interventions on collider parents, the strategy causal theory identifies as optimal; the non-stationarity argument for preference learning is recast as a *local, slow-drift* invariance matching the rank stability we measure (Section 3.5).

2 Related Work

Causal foundations. Structural causal models and do-calculus (Pearl, 2009, 1995) give the formal language for interventions. Observational data identifies only a Markov equivalence class (Spirtes et al., 2000); Reichenbach’s common-cause principle (Reichenbach, 1956) says why. Interventional data is therefore necessary to identify mechanisms (Peters et al., 2017) — the Rung-2 setting ACE operates in — where $n - 1$ interventions suffice and are sometimes necessary (Eberhardt et al., 2005; Eberhardt, 2006), greedy target selection is $(1 - 1/e)$ -optimal under submodular objectives (Shanmugam et al., 2015), and adaptive designs

need strictly fewer interventions than non-adaptive ones (Hauser and Bühlmann, 2012; Cho et al., 2016).

Bayesian active causal discovery. ABCI (Toth et al., 2022) and CBED (Tigas et al., 2022) maintain graph posteriors and design experiments by expected information gain, with multi-fidelity (Zhang et al., 2023) and sample-efficient (Zhou et al., 2024) extensions; GO-CBED (Zhang et al., 2025) targets a user-specified causal query. All target *structure discovery* and require posterior maintenance that is super-exponential in $|V|$ (Chickering, 1996). ACE addresses the complementary problem of *mechanism estimation* given known structure; skipping graph posteriors is what lets it run at 30 nodes, where these methods’ published evaluations do not yet reach, and it is also why our Bayesian-OED baseline’s per-decision oracle cost is so high (Section 5) — intrinsic to Monte Carlo posterior acquisition, not an implementation artifact.

Language models as priors and decision-makers. Pretrained LMs encode soft priors over world dynamics (Hao et al., 2023; Xie et al., 2022) and show non-trivial causal reasoning when prompted (Kiciman et al., 2024), though benchmarks find they struggle with long-horizon sequential experimental decisions (Zheng et al., 2025; Gandhi et al., 2025; Chen et al., 2025). A parallel line trains sequence models to decide *in context*: DPT (Lee et al., 2023) supervises a transformer to predict optimal actions from interaction histories with regret guarantees, and ICPE (Russo et al., 2025) meta-trains one for active sequential hypothesis testing. Conversely, Krishnamurthy et al. (2024) find that untrained LLM agents do *not* robustly explore without substantial intervention. ACE sits between these: it uses a sequence model as proposer but starts from natural-language pretraining rather than a curated task distribution, and treats the raw prior as necessary-but-insufficient, supplying lookahead-scored selection as the missing exploration structure; Section 4.2 quantifies where that structure is and is not load-bearing.

Preference optimization and learned discovery. DPO (Rafailov et al., 2023) learns from pairwise comparisons rather than reward magnitudes; active preference learning (Muldrew et al., 2024) extends it to selective acquisition. Concurrent causal-DPO work targets generative alignment, not experimental design: CAPO (Hu, 2026) reweights the DPO loss by an uncertainty decomposition and CDPO (Le et al., 2026) applies backdoor adjustment to confounded annotator preferences. ACE differs in problem, action space, and preference source (lookahead reward, not human annotation), and treats DPO as one calibration rule among several it compares (Section 3.4). Learned discovery via differentiable (Lorch et al., 2021) or RL-

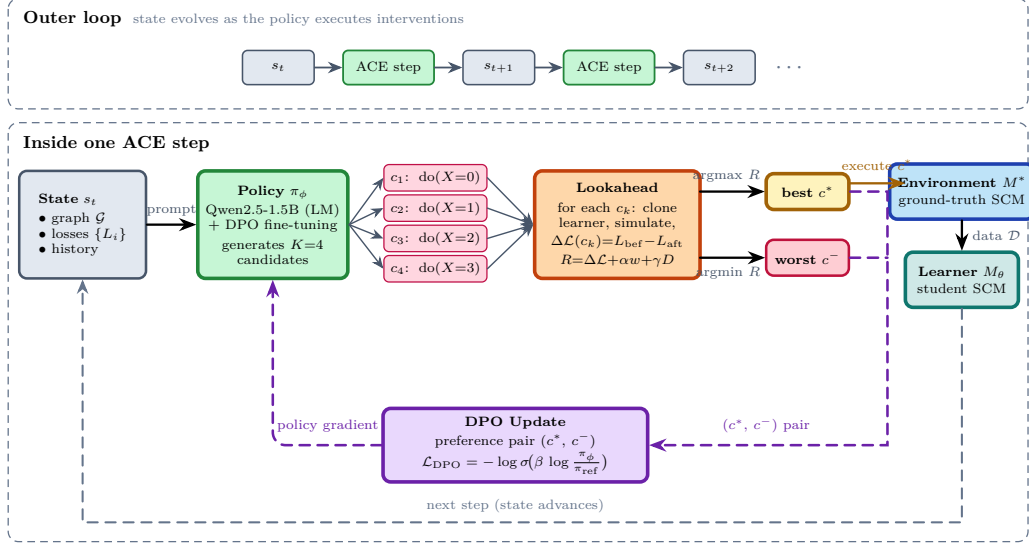


Figure 1: **ACE in one figure.** The state s_t (graph $\mathcal{G}$, per-node losses $\{L_i\}$, recent intervention history) is rendered as a text prompt. The LM policy π_ϕ (Qwen2.5-1.5B) generates $K=4$ candidates; lookahead clones the learner M_θ and scores each by composite reward $R = \Delta\mathcal{L} + w + \gamma D$. The argmax candidate c^* is executed on the environment M^* and updates the learner; the (argmax, argmin) pair (c^*, c^-) forms the DPO preference. Dashed arrows denote feedback; the lookahead arrow is exactly the environment-query cost audited in Section 5.

based (Sauter et al., 2024; Mukherjee et al., 2024) methods targets structure recovery and faces the same non-stationary rewards that motivate preference-based alternatives to value-based RL.

3 Method

3.1 Problem Setup and Preference Source

Formal setting (Rung 2). We adopt Pearl’s SCM framework (Pearl, 2009, 1995): $\mathcal{M} = \langle \mathcal{U}, \mathcal{V}, \mathcal{F}, P(\mathcal{U}) \rangle$ with endogenous $\mathcal{V} = \{V_1, \dots, V_n\}$ and structural equations $V_i = f_i(\text{Pa}_i, U_i)$. An intervention $\text{do}(V_i = \nu)$ replaces f_i with the constant ν , removing incoming edges to V_i . Because observational data cannot identify $\{f_i\}$ beyond the graph’s Markov equivalence class (Spirtes et al., 2000; Peters et al., 2017), ACE operates exclusively in the interventional regime. We take the graph $\mathcal{G}$ as known — the common case when domain experts supply the causal diagram — and target *mechanism estimation*: recovering $\{f_i\}$ from sequential experiments. Two components interact. The *environment* M^* is the ground-truth SCM; queried with $c = \text{do}(V_i = \nu)$ it returns samples from $P_{M^*}(\mathcal{V} \mid c)$, and every such query — whether executed against the real system or used only to score a candidate — counts as an environment query in the sense of Section 5. The *learner* M_θ maintains parametric mechanism estimates given $\mathcal{G}$, minimizing per-node losses $L_i = \mathbb{E}_c \|f_i^{M^*}(\text{Pa}_i, U_i) - f_i^{M_\theta}(\text{Pa}_i, U_i)\|^2$; reported total loss is $\sum_i L_i$.

Where preferences come from. They are not human-provided. Among the K candidates at each step, the highest-reward is labeled preferred over the lowest-reward, yielding one pair per step. The composite reward $R(c, s)$ acts purely as a preference oracle — it ranks candidates and emits the ordering, and its scalar magnitudes are then discarded. No value function or learned reward model is trained. This eliminates annotation cost and produces preference data at the same rate as experimental data.

3.2 Interaction Loop

The policy $\pi_\phi(c_t \mid s_t)$ observes $s_t = (M_\theta, \{L_i\})$ and proposes $c_t := \text{do}(V_i = \nu)$, with V_i ranging over $\mathcal{V}$ and ν drawn from a bounded interval fixed by the environment.

Candidate generation. The policy is Qwen2.5-1.5B (Qwen, 2024), prompted with a structured text encoding of (i) per-node losses $\{L_i\}$, (ii) parent sets per node, and (iii) recent intervention history. It generates K candidates as structured text (e.g. “do(X2 = 1.3)”), each parsed into a target and value; $K=4$ for the first 20 episodes and $K=3$ thereafter, a fixed compute-saving schedule rather than a tuned quantity.

Lookahead evaluation. Each candidate is scored by simulating its effect on a *cloned* learner. By default the clone receives data from executing c_k on the environment, and the loss reduction $\Delta\mathcal{L}(c_k) = L_{\text{before}} - L_{\text{after}}$ estimates information gain without committing — this

is precisely the step consuming K environment queries per step beyond the executed intervention, instrumented in Section 5. The `-lookahead_on_student` mode instead scores candidates against the learner’s own current beliefs, consuming zero environment queries during scoring, at the cost of ranking by a possibly-wrong internal model rather than ground truth. The winner $c^* = \arg \max_{c_k} R(c_k, s)$ is executed and M_θ updated; $(c_{\text{best}}, c_{\text{worst}})$ forms the preference pair.

3.3 Policy as Contextual Forward Model

Rung-2 design requires a forward model — the policy must anticipate what an intervention will do — and a policy is bounded by the quality of that prior. An MLP trained from scratch starts uniform; a one-step Bayesian acquisition function has a correct but shallow prior; a human experimentalist brings knowledge of how interventions propagate. A pretrained LM is the closest computational analogue: its corpus is largely about causal claims, so its weights encode a soft distribution over how variables relate (Hao et al., 2023; Kiciman et al., 2024), and in-context learning corresponds to implicit Bayesian inference under that prior (Xie et al., 2022). That prior alone is insufficient for reliable sequential exploration — Krishnamurthy et al. (2024) document exactly this failure in native LLM agents — which is what motivates the lookahead-scored selection step. Token generation also fits preference calibration: actions are text, so larger SCMs only lengthen the prompt, and the autoregressive factorization makes the log-ratios DPO consumes exact rather than approximated. Appendix A develops this framing in full.

3.4 Reward, Diversity, and Calibration Rules

The reward signal that generates preferences combines three components:

$$R(c, s) = \underbrace{\Delta \mathcal{L}}_{\text{info. gain}} + \underbrace{w(V_i, \{L_j\})}_{\text{importance}} + \underbrace{\gamma D(V_i, H)}_{\text{diversity}}, \quad (1)$$

where $\Delta \mathcal{L}$ is the loss reduction estimated by lookahead and $\gamma = 0.3$ is the diversity weight. We give w and D in full below; the constants are those used for every result in Section 4, verified against the released implementation.

Node importance. Let $\text{Ch}(V_i)$ be the children of V_i in $\mathcal{G}$ and n_i the number of times V_i has already been

targeted in the current episode. Node importance is

$$w(V_i, \{L_j\}) = \beta_w \cdot \frac{\tilde{L}_i}{\sum_k L_k + \epsilon} \cdot \frac{1}{\sqrt{1 + n_i}}, \quad (2)$$

$$\tilde{L}_i = \frac{1}{|\text{Ch}(V_i)|} \sum_{V_j \in \text{Ch}(V_i)} \omega_j L_j,$$

where $\omega_j = 2$ if $|\text{Pa}_j| \geq 2$ (i.e. V_j is itself a collider) and $\omega_j = 1$ otherwise, and with scale constant $\beta_w = 60.0$ and $\epsilon = 10^{-8}$; $w(V_i, \cdot) = 0$ if V_i has no children. This rewards targeting parents of high-loss *children*, doubly so when the child is itself a collider, normalized by total loss and discounted by $(1 + n_i)^{-1/2}$ against repeated targeting within an episode.

Diversity. Let H be the most recent $\min(100, |H_{\text{all}}|)$ intervention targets; $D = 0$ until 20 interventions have occurred, and otherwise

$$D(V_i, H) = 50 \cdot \text{Ent}(H) + 200 \cdot \max(0, \frac{1}{|V|} - f_{V_i}(H)) - 150 \cdot \max(0, c(H) - \tau(V_i)) \quad (3)$$

whose three terms are an entropy bonus, an under-sampling bonus, and a concentration penalty respectively. Here $\text{Ent}(H)$ is the Shannon entropy of the empirical target distribution over H , $f_{V_i}(H)$ is V_i ’s frequency in H , and $c(H) = \max_j f_{V_j}(H)$. The threshold $\tau(V_i)$ is 0.4 by default and relaxes to 0.75 when V_i is a parent of a collider whose loss still exceeds 0.3, so the penalty does not fight the strategic collider-parent concentration of Section 4.1 (Appendix E.8).

Calibration rules. Given the pair (y_w, y_l) from Section 3.2, we compare three rules for updating π_ϕ toward the winner, all on the identical pipeline above, so any downstream difference is attributable to the rule alone:

- **DPO** (default), the standard objective (Rafailov et al., 2023)

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(\beta \Delta_\phi)], \quad (4)$$

$$\Delta_\phi = \log \frac{\pi_\phi(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \log \frac{\pi_\phi(y_l | x)}{\pi_{\text{ref}}(y_l | x)},$$

with $\beta = 0.1$ and reference policy π_{ref} updated every 25 episodes.

- **SFT-best:** supervised fine-tuning on y_w alone, $\mathcal{L}_{\text{SFT}} = -\log \pi_\phi(y_w | x)$, discarding y_l and the reference-policy term entirely.
- **Ranking:** a pairwise Bradley–Terry logistic loss on policy log-probabilities with no reference policy, $\mathcal{L}_{\text{rank}} = -\log \sigma(\beta [\log \pi_\phi(y_w | x) - \log \pi_\phi(y_l | x)])$.

The comparison isolates what preference-based calibration adds over its simpler ingredients (imitating the

winner; ranking without a KL anchor). The infrastructure is released (`-policy_update {dpo, sft_best, ranking}`) and the matched-seed comparison on the 5-node benchmark is in progress (Appendix C.6, Table 4); a tie would be informative about *where* preference learning’s value lies, not a failure of the method.

3.5 Why Preference Learning?

Information gain from sequential experimentation is non-stationary: we measure $\mathbb{E}[\Delta\mathcal{L}]$ falling $\sim 500\times$ over a run while the *candidate ranking* within adjacent episodes stays stable (Spearman $\rho > 0.85$ between episodes t and $t + 1$). A collapsing reward scale over a stable within-step ordering is what makes a preference rule natural: DPO’s objective depends only on a log-ratio difference between candidates scored at the same step, so it is invariant to the scale factor a value-based estimator must track. We formalize this as a local, slow-drift decomposition (Assumption D.1) with a scale-invariance statement (Proposition D.2) in Appendix D, presented as motivation rather than as a central claim: the regularity it formalizes is measured directly, and the component it motivates is the one our ablations find dispensable (Section 4.2). $\Delta\mathcal{L}$ is also a Monte Carlo estimate of the mutual information $I(\theta; \mathcal{D}_c \mid s)$, so ACE pursues the Bayesian OED objective (Toth et al., 2022) through policy learning, at a per-decision oracle cost (Section 5) far below explicit posterior estimation (derivation in Appendix D.3).

4 Experiments

We evaluate ACE in four settings: a synthetic 5-node benchmark for controlled comparison against all baselines, a 30-node hierarchical SCM for scalability, a matched cross-scale sweep at 15–50 nodes separating the LM prior from the DPO updates, and coupled Duffing oscillators for physics transfer. A Phillips Curve application is in Appendix E.6. Five baselines span the canonical comparison points for active mechanism estimation under a known graph: Random and Round-Robin (Fisher, 1935) as non-adaptive references, Max-Variance (Cohn et al., 1996; Gal and Ghahramani, 2016) for uncertainty-driven greedy acquisition, Bayesian OED (Monte Carlo expected information gain; Toth et al., 2022; Tigas et al., 2022; Zhou et al., 2024) as the strongest principled baseline, and PPO (Schulman et al., 2017) as the value-based RL alternative. Seed counts: on the 5-node benchmark all methods use 5 seeds except Bayesian OED at 3 (higher per-seed compute); on the 30-node benchmark the passive baselines and Bayesian OED use 5 seeds and ACE is reported per seed group (Table 2), with PPO omitted as its per-step compute does not scale

within our budget. Appendix E gives per-seed values and learning curves. **All numbers in this section use executed-intervention-only accounting**; Section 5 supplies the dual accounting and the per-method query multipliers this convention omits.

4.1 Synthetic 5-Node Benchmark: ACE vs. Five Baselines

The 5-node benchmark (Figure 4; equations and learner in Appendix C) is a diagnostic testbed, not the headline: it isolates collider identification (X_3 with parents X_1, X_2), mixed functional forms, and a disconnected component ($X_4 \rightarrow X_5$) so that baseline behavior and ACE’s strategy can be read per node; the 30-node and Duffing results supply the scaling and transfer evidence.

Table 1 reports the headline result. The metric is total mechanism MSE summed across the five nodes (lower is better). The most informative comparison is against **Bayesian OED**, which receives the same correct graph and selects interventions by maximizing expected information gain via Monte Carlo posterior simulation. It is the strongest principled baseline available for the mechanism estimation setting. Even this principled method plateaus at a median of 1.99 (2.04 ± 0.12 mean), while ACE reaches a median of 0.61, a 69% improvement that isolates the contribution of *learned* preference-based policy optimization over greedy Bayesian acquisition — and, as the rightmost column previews and Section 5 quantifies, Bayesian OED’s per-decision oracle-query cost under total accounting is roughly two orders of magnitude larger than ACE’s, so this comparison does not favor ACE on hidden query cost. The traditional baselines (Random, Round-Robin, Max-Variance) plateau at median values between 2.09 and 2.21 even at 171 episodes, confirming that strategic adaptation, not additional data, drives improvement. PPO, given the same reward signal as ACE, fails to exploit it (median 2.06, mean 2.08 ± 0.06); we attribute this to the value-based estimator’s instability under non-stationary rewards (Section 3.5).

Strategic behavior. The policy puts 99.8% of interventions on the collider parents X_1, X_2 (vs. 40% under uniform sampling; 99.6–99.9% across seeds), cutting L_{X_3} from 3.3 to 0.054 (Appendix E.8).

4.2 Scaling: 30-Node Hierarchical SCM

To test scalability we evaluate on a hierarchical 30-node SCM (Figure 2) representative of the multi-layer structure found in gene regulatory and metabolic networks.

Table 1: Main results on the 5-node benchmark under executed-intervention-only accounting (Section 5 gives each method’s total-environment-query multiplier). ACE (N=5) runs until convergence (avg. 171 ep.); baselines (N=5) run at the matched budget. Bayesian OED: N=3. Improvement vs. ACE median (0.61); seed 789 is an outlier (quadratic-mechanism failure) justifying median over mean. p : Welch’s t , ACE vs. each baseline, unpaired (different seed sets), uncorrected; at $N=5$ a rank test cannot reach conventional significance after Holm correction. A paired replication of ACE on the baselines’ own seeds gives Holm-corrected paired- t $p \leq 0.026$ against all four heuristics (Appendix E).

Method	Mean $\pm$ Std	Median	Impr. vs. ACE	Welch p	Query mult.
ACE (171 ep., DPO)	0.92$\pm$0.73	0.61	—	—	4–5 $\times$
<i>Baselines at equal executed-intervention budget (171 ep.):</i>					
Random (Settles, 2009)	2.17 $\pm$ 0.07	2.21	72%	0.018	1 $\times$
Round-Robin (Fisher, 1935)	2.10 $\pm$ 0.11	2.09	71%	0.021	1 $\times$
Max-Variance (Cohn et al., 1996)	2.11 $\pm$ 0.14	2.10	71%	0.020	65 $\times$
Bayesian OED (Toth et al., 2022)	2.04 $\pm$ 0.12	1.99	69%	0.024	$\sim$ 500 $\times$
PPO (Schulman et al., 2017)	2.08 $\pm$ 0.06	2.06	70%	0.023	1 $\times$

Table 2: 30-node hierarchical SCM under *episode-matched* accounting. All methods use the same per-node MLP student learner; metric is total mechanism MSE. We report *best* total MSE over all training steps (step-level minimum) and *final* MSE at the last episode; Section 5 instead reports minima of *per-episode means*, a smoother and uniformly higher statistic, so its per-node numbers are not directly comparable to this table. Static baselines plateau within a few episodes (best $\approx$ final); ACE is non-monotone, so best-MSE is the cleaner converged summary. **ACE’s result at this scale is seed-dependent and we report both seed groups separately rather than pooling them behind a single mean.** The first three seeds are those of the original submission; the five added for this version do not reproduce them and instead sit at the passive-baseline plateau, for reasons traced to preference-optimization instability (see text). The all-seed row gives the exact pooled mean; we omit a pooled standard deviation because the distribution is bimodal and a single $\pm$ would misrepresent it. **The load-bearing 30-node comparison in this paper is the total-query-matched one in Section 5, not this table**, which we retain for continuity with the previous submission’s protocol.

Method	Best MSE	Final MSE	N	Improvement (best-vs-best)
ACE, seeds 42/123/456	1.95 $\pm$ 0.77	4.13 $\pm$ 3.61	3	3.0 $\times$
ACE, seeds 2022–2026	5.69 $\pm$ 0.69	10.72 $\pm$ 5.98	5	1.0 $\times$
ACE, all seeds	4.28	—	8	1.4 $\times$
Random	5.80 $\pm$ 0.05	5.85 $\pm$ 0.04	5	—
Round-Robin	5.79 $\pm$ 0.06	5.86 $\pm$ 0.07	5	—
Max-Variance	5.80 $\pm$ 0.04	5.86 $\pm$ 0.08	5	—
Bayesian OED	5.78 $\pm$ 0.05	5.90 $\pm$ 0.07	5	—

ACE generalizes to larger structures without architectural modification. All four baselines converge to a common plateau near 5.86 (stds < 0.08); at this scale differences among static heuristics are negligible, so what matters is whether a strategy adapts. The Bayesian-OED row is new to this version, answering a reviewer’s objection that the previous submission asserted OED does not scale without showing it: the faster Monte Carlo variant of Appendix C.6 does run at 30 nodes and lands within noise of the passive heuristics (5.78 $\pm$ 0.05) despite its far higher per-decision oracle cost — itself the point of Section 5.

The episode-matched ACE result at this scale does not replicate across seeds. Five further seeds under the identical configuration and budget yield 5.69 $\pm$ 0.69 against the original three seeds’ 1.95 $\pm$ 0.77

— indistinguishable from the passive baselines, for an all-seed mean of 4.28. The runs are mechanically sound (full 300 episodes, single job, no resume); what separates them is preference-optimization stability: all five show pathological training diagnostics (two at chance-level DPO loss, three collapsed to near-zero loss with extreme reference divergence) and all five land at the plateau. We read this as fragility, not noise: *when the preference update destabilizes, the episode-matched advantage disappears*, and the original three seeds are the subset in which it did not. Two consequences follow. Section 5’s total-query-matched comparison, a within-batch measurement whose ACE runs show healthy diagnostics throughout, therefore carries the 30-node claim; and Contribution 1 is sharpened rather than softened, since the component whose instability

erases the advantage is the one whose removal leaves performance intact.

How much is the LM prior versus DPO? We compare ACE against *ACE without DPO* — the same proposer and lookahead with weight updates disabled (distinct from the random-proposer ablation of Table 3, which removes the LM too). On the canonical 30-node SCM the two are not distinguishable ($\text{ACE } 1.95 \pm 0.77$ vs. 1.67 ± 0.19 , step-level best as in Table 2; Welch $p = 0.59$, Mann-Whitney $p = 1.0$, at $N=3$ vs. 5 — too few seeds for an equivalence test), and under *anonymization* of node names (abstract `n_xxxx` tokens) they are equivalent in the formal sense: ACE 1.51 ± 0.16 vs. 1.49 ± 0.18 ($N=5$, paired by seed; paired t $p = 0.37$, Wilcoxon $p = 0.81$, TOST equivalence within ± 0.15 total MSE at $p = 0.001$). This is Contribution 1: the pretrained prior plus lookahead carries the load at this scale. The seed expansion of Table 2 pushes the conclusion further in the same direction — across all eight canonical seeds full ACE averages 4.28 while ACE-w/o-DPO, with no preference update to destabilize, shows no comparable failure — so we state the claim as: disabling the preference update costs no measurable performance, and leaving it enabled carries a stability risk the anonymized and matched-budget runs do not exhibit. Learning curves are in Appendix E.3 (Figure 6).

Scaling across graph size. A matched sweep on the hierarchical LargeScaleSCM family (Figure 5, Appendix E.3) extends this at $N \in \{15, 30, 50\}$ ($N=5$ seeds per cell, fixed ~ 40 -episode plateau budget). ACE and ACE-w/o-DPO form a band (0.051–0.136 per node) well below Random (0.153–0.192) at every size, so adaptive LM reasoning scales while the static heuristic degrades. Within the band, however, DPO does not help and at two sizes hurts: ACE-w/o-DPO is better per node at $N=15$ (0.051 vs. 0.062; paired t , $p = 0.022$) and at $N=30$ (0.067 vs. 0.114; $p = 0.008$), and the two are not distinguishable at $N=50$ (0.136 vs. 0.118; $p = 0.29$) — the same direction as the seed-expansion finding above. A lookahead-breadth ablation at $N=50$ ($K \in \{4, 8, 16\}$) finds best-MSE flat in K while per-episode oracle cost rises linearly (Section 5), pointing to salience-targeted candidate generation rather than breadth as the lever at larger N ; details in Appendix E.3.

4.3 Component Ablation

The diversity reward is the most critical component (Table 3; per-seed detail in Appendix E.4): without it the policy collapses onto a single node and *worsens* beyond baseline (2.82). Replacing the LM proposer with random proposals — which removes the DPO update with it — matches Round-Robin (2.10), so learned propo-

posal generation, not lookahead, drives the improvement; that row is not a DPO ablation, and the DPO-only comparison that speaks to Contribution 1 is the one in Section 4.2. The node-importance row ($\beta_w = 0$, `-cov_bonus 0.0`) is queued, not omitted.

4.4 Physics: Coupled Duffing Oscillators

We apply ACE to three coupled Duffing oscillators (Kovic and Brennan, 2011) with true coupling $X_1 \leftrightarrow X_2 \leftrightarrow X_3$ and a synchronization-induced spurious X_1-X_3 correlation. Coupling-parameter error is 0.042 ± 0.036 for ACE versus 0.245 ± 0.121 (Random) and 0.238 ± 0.076 (Round-Robin) ($N=5$, 100 episodes; Appendix E.5). The policy concentrates 62% of interventions on X_2 , the oscillator whose clamping breaks the spurious correlation, without being told to.

5 Budget Accounting

An intervention budget can be counted two ways: interventions *executed* (what deployment would spend) or every environment query *consumed*, including those used only to score candidates. Every adaptive method in Table 1 uses the second kind internally — ACE’s lookahead, Max-Variance’s uncertainty probe, Bayesian OED’s Monte Carlo EIG estimate — so executed-only accounting, as in the previous version of this paper, understates training cost, and unequally across methods.

Instrumentation and multiplier audit. We tag every environment query as `executed`, `lookahead`, or `observational` in ACE and in an `InstrumentedOracle` wrapper around every baseline’s ground-truth SCM; each run writes a per-tag `query_budget.json` (Appendix C.6). Table 1’s last column reports each method’s ratio of total to executed queries, re-derived from the exact configuration that produced its row. Random, Round-Robin, and PPO never probe before executing: $1\times$. ACE’s lookahead scores $K \in \{3, 4\}$ candidates per step beyond the one executed: $4-5\times$. Max-Variance scores $n_{\text{candidates}} = 64$ per step by MC-dropout: $65\times$ — the same asymmetry raised against ACE, previously invisible because uninstrumented. Our Bayesian-OED baseline scores $|\mathcal{V}| \times 10$ candidate values with 20 posterior draws each before executing one: $\sim 500\times$ in raw samples (derivation in Appendix C.6), a cost intrinsic to Monte Carlo EIG estimation rather than to our implementation. The instrumentation also enables two things the previous version lacked: the `-lookahead_on_student` variant (exact $1\times$; candidates scored against the learner’s own model), and a `-query_budget` flag that stops training at a specified *total* sample count, so every method can be rerun at

exactly matched cost.

Matched-budget rerun, 30 nodes. With every baseline given ACE’s own environment-lookahead total (mean 483,460 samples, $N=5$ seeds) on the 30-node SCM of Table 2, ACE reaches per-node best MSE 0.110 ± 0.045 — 43% better than the best baseline (Random 0.192 ± 0.002 ; Round-Robin 0.193 ± 0.002 ; Max-Variance 0.194 ± 0.001 ; Bayesian OED 0.194 ± 0.002). The four baselines are indistinguishable at this budget, consistent with strategy mattering less as the query budget approaches the passive-sampling limit — exactly the regime where an adaptive advantage is hardest to keep. Every ACE seed individually beats the pooled baseline mean (worst 0.157 vs. 0.192). The `-lookahead_on_student` variant does better still: at its own honest budget of 130,140 samples, $3.7\times$ fewer than the baselines received, it reaches 0.096 ± 0.018 , a 50% margin.

Matched-budget rerun, 5 nodes. The identical protocol on Table 1’s benchmark (shared budget 3,204,460 samples; Bayesian OED compared separately under executed-only accounting at this scale, Appendix C.6) gives a larger margin: ACE 0.094 ± 0.004 per node against a best baseline of 0.396 ± 0.028 (PPO; the others 0.400–0.402), a 76% improvement, and the student-lookahead variant 0.096 ± 0.001 at 615,680 samples, $5.2\times$ fewer than the baselines. Every ACE seed again beats the pooled baseline mean (worst 0.098 vs. 0.396). The decision gate is resolved in ACE’s favor at both scales, under both accountings, with the smaller graph showing the larger margin.

Checkpoint selection. We report best-episode MSE throughout this section. ACE’s best and final MSE diverge in a way the baselines’ do not: 0.110 vs. 0.276 at 30 nodes ($2.5\times$) and 0.094 vs. 0.171 at 5 ($1.8\times$) — an early plateau followed by drift — whereas every baseline’s final matches its best within 1–4%. Under the strictest reading, ACE’s *final* against each baseline’s *best*, ACE still wins at 5 nodes (0.171 vs. 0.396) but not at 30 (0.276 vs. 0.192). The 30-node margin therefore depends on checkpoint selection; the 5-node margin does not. Where no validation signal for early stopping exists, the student-lookahead variant, whose best-to-final gap is the smaller at both scales ($1.5\times$ and $2.2\times$), is the configuration we recommend.

6 Limitations and Future Work

ACE addresses *mechanism estimation given known structure* — the setting whenever a graph comes from domain expertise, prior studies, or an upstream structure learner. Joint structure-and-mechanism methods (ABCI, CBED, GO-CBED) target a different problem in a different computational regime, since graph poste-

riors are NP-hard (Chickering, 1996) and intractable beyond roughly twenty variables (Zhou et al., 2024); ACE complements rather than competes with them. The binding cost in our setup is prompt length, $O(N+E+H)$ in the default encoding; the compact salience-ranked encoding used at 50 nodes and the plateau-budgeted protocol that follows from it are given in Appendix E.3, with the 100+ regime, a structure-learning front end, and graph misspecification (Appendix F.2) as the natural follow-ups.

Two caveats on the matched-budget result. First, at 30 nodes — though not at 5 — the margin does not survive the strictest final-vs-best reading (Section 5). Second, the `sft_best` and `ranking` calibration-rule comparison of Section 3.4 is released but not yet reported (Appendix C.6, Table 4); per our own framing, a tie there would be informative rather than grounds to withdraw a claim.

Preference-optimization stability at 30 nodes is unresolved and is our most important open problem. Five of five newly added canonical seeds show pathological DPO training and land at the passive-baseline plateau (Section 4.2); we have characterized this failure but not diagnosed what separates it from the cleanly converging runs behind Section 5’s result. Until that is understood, we recommend the configurations that do not depend on the preference update — ACE-w/o-DPO and `-lookahead_on_student` — and say so in the main text because it bears directly on how much of the method a practitioner should adopt.

7 Conclusion

ACE learns sequential experimental design for causal mechanism estimation from a pretrained LM proposer, a lookahead step scoring candidates against a cloned learner, and a preference-based calibration rule. Disentangling these is the paper’s central move: at 30 nodes, under anonymization, and across a 15–50-node sweep, the proposer and lookahead alone match or exceed the DPO-calibrated method, and expanding the canonical 30-node comparison to eight seeds shows the preference update to be a source of instability rather than of advantage. ACE leads five baselines under conventional executed-intervention accounting and, under the total-query accounting we instrument for every method, wins at matched cost at both scales, with the hidden-cost concern falling more heavily on two of our own baselines than on ACE. A contribution should be stated at the granularity its own ablations support: here, the LM-prior-plus-lookahead mechanism, with preference calibration a secondary component still being established against simpler alternatives.

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- ## Checklist
1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] Section 3 and Appendix C.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes] The per-decision oracle-query complexity of every method is derived in Section 5 and Appendix C.6; prompt-length scaling and per-episode wallclock are in Appendix E.3.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes] Anonymized repository linked in Appendix G.
 2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes] Assumption D.1, stated in Appendix D and summarized in Section 3.5.
 - (b) Complete proofs of all theoretical results. [Yes] Appendix D.1. We note in Section 3.5 that Proposition D.2 is presented as a motivating illustration rather than a load-bearing contribution.
 - (c) Clear explanations of any assumptions. [Yes] Section 3.5, with the supporting rank-stability evidence reported alongside.
 3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes] Appendix G; every experiment’s exact job script is named in Appendix C.6.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes] Appendix C and the hyperparameter table therein.

- (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes] All results report mean $\pm$ standard deviation across seeds, with seed counts stated per table. Where a distribution is bimodal across seeds we report the groups separately rather than a single mean (Table 2).
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] Appendix G.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
- (a) Citations of the creator If your work uses existing assets. [Yes] The pretrained policy model and all baseline algorithms are cited at first use.
 - (b) The license information of the assets, if applicable. [Yes] Appendix G.
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes] Anonymized repository, Appendix G.
 - (d) Information about consent from data providers/curators. [Not Applicable] All environments are simulated or drawn from public macroeconomic series.
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable] No personally identifiable or offensive content is involved.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
- (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

ACE: Language-Model Priors and Lookahead Selection for Active Causal Mechanism Estimation: Supplementary Materials

A The Role of World Models in Causal Experimental Design

The choice of a pretrained language model as ACE’s policy is not a matter of engineering convenience. It reflects a specific epistemological position about what is required to plan good experiments in causal domains, and about what kind of prior knowledge a policy must bring. This appendix develops that argument in full.

A.1 Experimental Design Requires a Forward Model

Pearl’s causal hierarchy (Pearl, 2009; Pearl and Mackenzie, 2018) distinguishes three rungs of causal knowledge: association (Rung 1), intervention (Rung 2), and counterfactual reasoning (Rung 3). Active experimental design operates at Rung 2. The question an experimentalist must answer at every step is not “what do the data say?” but “what would happen if I intervened on V_i with value ν ?” Answering this question requires a forward model of the system that can simulate the downstream effects of hypothetical interventions.

Crucially, the quality of the policy is bounded above by the quality of its forward model. An agent with a perfect forward model can compute expected information gain exactly and pick the optimal intervention. An agent with a uniform forward-model prior must rely entirely on observed data, episode by episode, to discover which interventions reveal informative causal structure. For any fixed intervention budget, there is a direct tradeoff: stronger prior forward knowledge permits fewer experiments.

ACE’s student SCM is an explicit forward model of the system, parameterized by per-node MLPs. But it begins training with random weights, which means it embodies only the weakest possible prior: that the mechanisms are whatever an MLP can represent. This is much weaker than what a human experimentalist brings to the same problem. A physicist intervening on a coupled oscillator chain already knows that clamping the middle oscillator decouples the endpoints. A molecular biologist designing a gene-knockout campaign already knows that transcription factors with many targets are high-value intervention points. An economist studying inflation dynamics already knows that monetary tightening episodes expose nonlinearities absent during stable regimes. These priors are not learned from experiment; they are brought to experiment.

A.2 Language Models Encode Causal Priors Through the Structure of Scientific Text

A language model pretrained on a corpus that includes scientific articles, textbooks, experimental reports, and technical derivations encodes, in its weights, a soft distribution over how variables in real systems relate causally. This encoding is not incidental but structural: scientific text is in the business of articulating causal claims. A textbook paragraph describing a chain of coupled oscillators contains sentences of the form “an intervention on the middle oscillator decouples the endpoints,” “the coupling parameter κ mediates synchronization,” and “clamping X_2 breaks the spurious correlation between X_1 and X_3 .” Each such sentence is an interventional claim (Rung 2), and each increases the language model’s log-probability of generating similar interventional proposals when prompted with similar context.

Kiciman et al. (2024) demonstrate this empirically. They evaluate GPT-4 on pairwise causal direction queries across multiple benchmarks and find substantially above-chance performance, with the model correctly identifying the causal direction from contextual cues that are linguistic in origin (e.g., descriptions of temporal order, physical mechanism, domain-typical relationships). The LM has not learned causality through experimentation; it has absorbed the linguistic regularities under which causal statements are made in scientific text, and those regularities track real causal structure because they are produced by writers who understand that structure.

Hao et al. (2023) extend this observation into a framework. They frame language model reasoning as planning in a world model, where the LM serves simultaneously as the reasoner (the system generating action proposals) and the world model (the system predicting the outcomes of those actions). This dual role is exactly what ACE requires at each step: a policy that both proposes candidate interventions and has some implicit model of how those interventions will propagate through the system. The LM unifies these two functions in a single pretrained artifact.

Xie et al. (2022) formalize the connection to Bayesian inference. They show that in-context learning corresponds to implicit Bayesian posterior update under the pretraining prior: given a prompt describing the current epistemic state (graph structure, per-node losses, recent interventions), the LM’s next-token distribution approximates the Bayesian predictive distribution over continuations, conditioned on both the prompt and the pretrained prior. ACE exploits this machinery: the prompt at step t encodes everything the agent has learned about the specific system, and the LM’s generation is an approximate posterior predictive distribution over sensible interventions given that state.

A.3 Why Not a Fixed-Architecture MLP or Hand-Coded Heuristic?

Two obvious alternatives exist, and each forgoes the forward-model prior.

An MLP policy trained from scratch on a single SCM inherits no causal prior. It must learn from experience that, for example, collider parents are high-value intervention targets. In our 5-node experiments, this is observable: random policies take on the order of 100 episodes to reach the baseline plateau of 2.10, while ACE’s LM-based policy reaches a median of 0.61 in fewer episodes. The gap is not because the LM is learning faster; it is because the LM starts from a prior that already weakly favors strategic interventions, and DPO refines that prior rather than constructing it from nothing. An MLP at the same parameter count would have to both construct the prior and fit the system, within a fixed intervention budget.

A hand-coded heuristic such as greedy expected information gain (Bayesian OED) does bring a prior, but it is the wrong kind. Greedy EIG is a one-step-lookahead acquisition function; it cannot reason about multi-step strategies like “first resolve the identification of X_2 to unlock information about the X_3 collider.” In our 5-node results, Bayesian OED plateaus at 2.04 ± 0.12 , decisively below ACE at median 0.61. The greedy prior is correct but shallow.

A pretrained LM brings a qualitatively different kind of prior: one that is deep enough to encode multi-step experimental intuitions (because those intuitions appear in the training corpus as whole-paragraph descriptions of experimental strategy), but one that must be calibrated to the specific system. DPO performs that calibration. The LM without DPO is a broad but uncalibrated generator; DPO without a pretrained prior would be an empty calibrator. Together they form a policy that has both breadth and specificity.

A.4 Empirical Signatures of the Forward-Model Prior in ACE

Three empirical observations in the main paper are direct consequences of the forward-model framing.

Transfer across domains. The same LM weights produce sensible candidates on 5-node synthetic SCMs, 30-node hierarchical SCMs, coupled Duffing oscillators, and economic time series. No architectural change is needed across these domains. A fixed-architecture MLP would require re-training per domain; a hand-coded heuristic would require per-domain re-tuning. The LM’s transfer is a signature of a domain-agnostic world-model prior being specialized by prompt context, in line with Xie et al. (2022)’s in-context-learning-as-Bayesian-inference account.

Emergent collider concentration without explicit instruction. ACE’s 99.8% concentration on collider parents is not something the policy was told to do. It emerges from preference optimization. The emergence is consistent with the LM having a prior that already weakly favors such concentrated interventions in collider contexts (because scientific text describing collider identification contains such prescriptions), with DPO sharpening that weak prior into a strong preference through observation of reward signal.

Baseline convergence on 30-node. Random, Round-Robin, and Max-Variance all converge to the same plateau at ~ 5.86 on 30 nodes. This indicates that at this scale, no static heuristic can resolve the mechanism estimation problem within budget. What distinguishes ACE is not parameter count or compute but prior knowledge: ACE

brings the LM’s distribution over plausible interventions into a regime where purely data-driven methods stall.

A.5 What the Forward-Model Framing Does Not Claim

To be precise: we do not claim the LM is a correct or complete causal reasoner. Gandhi et al. (2025); Chen et al. (2025) show that LLMs struggle with long-horizon sequential experimental decisions in zero-shot settings, and Zheng et al. (2025) survey the limitations of LLM causal discovery. Our claim is narrower. We claim that (i) a pretrained LM encodes a non-trivial forward-model prior, (ii) this prior is a better starting point than a uniform prior or a one-step-lookahead heuristic, and (iii) DPO provides the calibration machinery to specialize this prior to any specific system. ACE is best understood as a method for combining broad pretrained causal knowledge with narrow, calibrated preference learning, rather than as a claim that LMs alone can solve causal experimental design.

B Notation and Problem Formulation (Extended)

We adopt Pearl’s causal framework (Pearl, 2009, 1995). A Structural Causal Model (SCM) $\mathcal{M} = \langle \mathcal{U}, \mathcal{V}, \mathcal{F}, P(\mathcal{U}) \rangle$ consists of exogenous variables $\mathcal{U} = \{U_1, \dots, U_m\}$, endogenous variables $\mathcal{V} = \{V_1, \dots, V_n\}$, structural equations $\mathcal{F} = \{f_1, \dots, f_n\}$ where $V_i = f_i(\text{Pa}_i, U_i)$, and a distribution $P(\mathcal{U})$.

The causal relationships induce a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $(V_j, V_i) \in \mathcal{E}$ iff $V_j \in \text{Pa}_i$. The observational distribution factorizes as

$$P(V_1, \dots, V_n) = \prod_{i=1}^n P(V_i \mid \text{Pa}_i). \quad (5)$$

An intervention $\text{do}(V_i = \nu)$ replaces f_i with the constant ν and removes all edges into V_i in the mutilated graph $\mathcal{G}_{\overline{V_i}}$ (Pearl, 2009). This is a Rung-2 operation in Pearl’s causal hierarchy: it requires active manipulation of the system rather than mere observation, and yields samples from the post-intervention distribution $P(V_j \mid \text{do}(V_i = \nu))$ for all $j \neq i$.

The impossibility of mechanism identification from observational data alone follows from Reichenbach’s common-cause principle (Reichenbach, 1956): for any bivariate association $A \not\perp\!\!\!\perp B$, there exist at least three causal structures ($A \rightarrow B$, $B \rightarrow A$, $A \leftarrow C \rightarrow B$) that are Markov equivalent (Spirtes et al., 2000; Peters et al., 2017). Distinguishing structural equations f_i requires interventional data (Peters et al., 2017).

ACE’s task. Given known $\mathcal{G}$, ACE learns the structural equations $\{f_i\}$ by sequentially selecting interventions $c_t = \text{do}(V_{i_t} = \nu_t)$ to minimize the total mechanism estimation loss. The policy $\pi_\phi(c_t \mid s_t)$ adapts based on observed losses $\{L_i\} = \{L_1, \dots, L_n\}$ where L_i measures current discrepancy between the student SCM mechanism $\hat{f}_i$ and the true f_i .

C Implementation Details

C.1 Ground Truth SCMs

5-node benchmark. The benchmark used throughout the main paper has structural equations

$$\begin{aligned} X_1 &\sim \mathcal{N}(0, 1) && \text{(exogenous root)} \\ X_4 &\sim \mathcal{N}(2, 1) && \text{(exogenous root)} \\ X_2 &= 2X_1 + 1 + \varepsilon_2 && \text{(linear)} \\ X_3 &= 0.5X_1 - X_2 + \sin(X_2) + \varepsilon_3 && \text{(nonlinear collider)} \\ X_5 &= 0.2X_4^2 + \varepsilon_5 && \text{(quadratic)} \end{aligned}$$

with $\varepsilon_i \sim \mathcal{N}(0, 0.01)$. The graph has two roots (X_1, X_4), two intermediate nodes (X_2, X_5), and one collider (X_3 with parents X_1, X_2). The collider tests the policy’s ability to identify the nonlinear interaction term $\sin(X_2)$, which requires observing X_3 under targeted interventions on both X_1 and X_2 . The disconnected chain $X_4 \rightarrow X_5$ with a quadratic mechanism tests mechanism estimation for polynomial functions.

30-node hierarchical SCM. Constructed at runtime with seed-controlled coefficients in $[0.3, 0.7]$. Five-layer structure (see Figure 2):

- Layer 0: 5 root nodes, $X_i \sim \mathcal{N}(0, 1)$.
- Layer 1 (X6–X10): 1–2 root parents each, linear combination with noise std 0.15.
- Layer 2 (X11–X20): Mix of single-parent and two-parent (collider) nodes; every third node is a collider.
- Layer 3 (X21–X25): Complex colliders with 2–3 parents from Layer 2; sine nonlinearity on every fifth node.
- Layer 4 (X26–X30): Single-parent leaf nodes.

The seed controls the random graph wiring within these constraints, ensuring reproducibility.

Duffing oscillators. Three coupled Duffing oscillators with true coupling $X_1 \leftrightarrow X_2 \leftrightarrow X_3$. The synchronization-induced correlation between X_1 and X_3 is a spurious association (Rung 1) with no direct causal edge; interventions on X_2 are necessary to break this correlation and estimate the direct couplings.

C.2 Learner Architecture

The student SCM uses a `ModuleDict` of per-node mechanisms:

- **Root nodes:** Learnable parameters $(\hat{\mu}, \hat{\sigma})$; loss is KL divergence between $\mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ and empirical samples.
- **Non-root nodes:** 2-layer MLP with hidden dim 64 and ReLU activations, taking the parent vector Pa_i as input and outputting a scalar prediction of V_i . Loss is MSE against observed V_i under the current intervention.

Optimization: Adam (Kingma and Ba, 2015), learning rate 2×10^{-3} , batch size 100. Per-step training runs 100 epochs on the latest interventional batch; observational training every 3 steps runs 100 epochs on 200 fresh observational samples to maintain root-distribution estimates.

C.3 Policy Architecture and Prompt Design

The policy π_ϕ is Qwen2.5-1.5B (Qwen, 2024), accessed via the HuggingFace Transformers API. Decoding for candidate proposals uses temperature 0.8, top- p 0.9, repetition penalty 1.3, and at most 64 new tokens. The prompt is *problem-first*: it leads with the failing mechanisms (not the graph), states explicitly which parents to intervene on, warns against over-sampled nodes, and ends at a decision cue so the model completes with a node choice rather than defaulting to the first node. A representative instance (5-node SCM):

```
PROBLEM: Node losses (high=bad): X3=1.45, X4=0.98, X1=0.92.
To fix X3, intervene on its parents: X1, X2.
Recent interventions: [X1:2/20, X2:14/20, ...]. WARNING: X2 is
over-sampled (14/20). AVOID X2.
Graph: [X1->X2, X1->X3, X2->X3, X4->X5].
Valid targets: X1, X2, X3, X4, X5. Value range: [-5.0, 5.0].
Examples of good reasoning: ...
Based on the current losses, which node should we intervene on
to learn the failing mechanism?
Reasoning: The highest loss node is
```

The five prompt blocks are: (i) a *loss ranking* of the top-3 failing non-root nodes with an explicit parent-intervention hint; (ii) a *history* summary with an over-sampling warning when any node exceeds 30% of recent interventions; (iii) the *graph* edge list; (iv) the *valid targets* and value range; and (v) *few-shot* reasoning exemplars followed by the decision cue. We generate $K = 4$ candidates per step by re-sampling, decaying to $K = 3$ after warm-up.

Full vs. compact encoding. The default (`-prompt_strategy full`) lists every node and every edge, so prompt length grows as $O(N+E+H)$ in nodes, edges, and history; this is the binding cost at scale (Section 6). The compact encoding (`-prompt_strategy compact`, with `-prompt_top_m`) instead surfaces only the top- m highest-loss non-root nodes and their parents, restricts the graph to edges incident to that salient set, and limits the candidate pool to those parents. This holds prompt length at $\sim O(m)$ independent of N (empirically $\sim 41\%$

shorter at 50 nodes for $m=8$) while preserving the same five-block structure; it is validated for best-MSE parity at $N=30$ before being used at $N=50$ and is mandatory for the 100+ regime.

Node anonymization. To isolate the LM’s pretrained semantic prior from its structural reasoning, node names $X_1 \dots X_N$ can be replaced by abstract hex tokens `n_xxxx` (deterministic in the seed). Under anonymization the few-shot exemplars switch from the hard-coded $X_1/X_2/X_3$ form to generic placeholders drawn from the current node set (so the model is never taught to emit invalid names), and a stable position index preserves the per-mechanism nonlinearity rule despite the renaming.

Treatment of the different modalities. A central design choice is that *every* domain is mapped onto the *same* text interface above; only the node vocabulary and the physical meaning of an intervention change, which is precisely why ACE needs no architectural modification across domains.

- **Synthetic SCMs (5-node; hierarchical 30/50-node).** Nodes $X_1 \dots X_N$; interventions $\text{do}(X_i=\nu)$ over $[-5, 5]$. The hierarchical family adds layered colliders; the anonymized variant strips semantics.
- **Physics — coupled Duffing oscillators.** Nodes are the oscillator displacements in a chain $X_1 \leftrightarrow X_2 \leftrightarrow X_3$; an intervention is a *clamp* $\text{do}(X_{\text{mid}}=0)$ that physically pins the intermediate mass and breaks the synchronization-induced spurious correlation. The mechanisms are ODE-derived but the prompt sees the same edge list and per-node losses.
- **Economics — US Phillips curve (FRED).** Nodes are real macroeconomic series (`UNRATE`, `FEDFUNDS`, `MICH`, `CPILFESL`); “intervention” is active selection of a historical data subset rather than a physical *do* (stated as such in the main text). Here the semantically meaningful names let the LM’s pretrained economic prior engage most strongly — the opposite end of the spectrum from the anonymized SCM control.

The anonymized-vs-named contrast across these modalities is what the LM-prior-vs-DPO analysis (Appendix E.3) exploits: performance is preserved even when names carry no meaning, indicating the structural-plus-loss reasoning, not name semantics alone, drives intervention choice.

C.4 Reward Components in Full

The reward in Equation (1) has three components, given in full in the main text (Eqs. 2–3, Section 3.4); this subsection adds implementation detail and corrects the previous submission’s appendix, which stated $\alpha = 0.1$, `cov_bonus` = 60.0, and $\gamma = 0.05$. Re-deriving these constants directly from the code path actually exercised by `jobs/hpc_ace_seed.sh` (the job that produced every ACE number in Table 1, which passes no reward-weight flags and therefore relies entirely on `ace_experiments.py`’s argparse defaults) shows no separate α multiplies the node-importance term: the term is $w = \text{cov_bonus} \cdot \text{node_weight} \cdot \text{under_sample}$ with `cov_bonus` = 60.0 acting as the term’s entire scale constant, and the diversity weight actually used is $\gamma = \text{diversity_reward_weight} = 0.3$, not 0.05. Eqs. 2–3 in the main text reflect the corrected values and are the ones used to produce every number in this paper.

Information gain. $\Delta\mathcal{L}(c) = L_{\text{before}} - L_{\text{after}}$ where $L = \sum_i L_i$ is the validation loss summed across nodes, estimated by lookahead on a cloned learner.

Node importance. Eq. 2 uses `node_weight` = $\text{direct_child_impact_weight}(\cdot) / (\sum_k L_k + \epsilon)$, i.e. the loss-weighted impact on V_i ’s *children* (doubled for children that are themselves colliders), not V_i ’s own loss L_i as an earlier draft of this appendix stated; this rewards intervening on parents of poorly-understood, structurally important children. `under_sample` = $(1 + n_i(H))^{-1/2}$ where $n_i(H)$ counts interventions on V_i within the current episode, preventing over-concentration on a single node.

Diversity. $D(V_i, H)$ (Eq. 3) combines an entropy bonus over the empirical target distribution in the trailing history window H ($|H| \leq 100$), an under-sampling bonus for nodes below their uniform-allocation share of H , and a concentration penalty once any node exceeds an adaptive threshold $\tau(V_i)$ of H (relaxed for active collider parents so the penalty does not fight the strategic concentration in Section E.8). It is realized in code as `compute_unified_diversity_score` and is gated to 0 until at least 20 interventions have occurred (avoiding noisy entropy estimates early in an episode).

These weights keep $\Delta\mathcal{L}$ typically the largest single term early in training, when losses (and hence achievable loss reductions) are largest, while γD provides the exploration pressure the component ablation (Table 3) shows is necessary: removing it ($\gamma = 0$) is the single most damaging ablation we test.

C.5 Bayesian OED Baseline Implementation

Our Bayesian OED baseline approximates the expected information gain over the same parametric mechanism class as ACE’s student. At each step:

1. For each candidate (V_i, ν) with ν uniformly sampled from $[-5, 5]$:
 - (a) Sample $M = 10$ posterior parameter draws $\theta^{(m)} \sim P(\theta \mid \mathcal{D}_{\text{seen}})$ by training M student copies from different random initializations.
 - (b) For each draw, simulate outcome $y^{(m)} \sim P(\cdot \mid \theta^{(m)}, \text{do}(V_i = \nu))$.
 - (c) Estimate $\widehat{\Delta\mathcal{L}} = \frac{1}{M} \sum_m [L(\theta^{(m)} \mid \mathcal{D}_{\text{seen}}) - L(\theta^{(m)} \mid \mathcal{D}_{\text{seen}} \cup \{(V_i, \nu, y^{(m)})\})]$.
2. Execute the candidate with highest $\widehat{\Delta\mathcal{L}}$.

We use $|\mathcal{V}| \times 10$ candidate values per step (ν discretized on a 10-point grid over $[-5, 5]$ via `np.linspace`), each scored with $M = 20$ posterior draws, i.e. $|\mathcal{V}| \times 10 \times 20 = 1,000$ oracle-scored candidates for $|\mathcal{V}| = 5$ (correcting the previous submission’s “ $|\mathcal{V}| \times 20$ candidates,” which described only the value grid times a mis-stated per-value draw count and did not carry through to the query-cost accounting in Appendix C.6). This faithfully adapts the ABCI (Toth et al., 2022) and CBED (Tigas et al., 2022) acquisition-function framework to the mechanism estimation setting where graph structure is known. Each Bayesian OED *decision* (the 1,000-candidate evaluation plus one executed intervention) takes ~ 4 –6 minutes on a CPU, limiting us to $N=3$ seeds; Appendix C.6 shows this cost is exactly the source of the baseline’s large environment-query multiplier.

C.6 Budget Accounting: Full Multiplier Derivation

Section 5 reports, for each method in Table 1, the ratio of total environment queries (every call to the ground-truth SCM’s `generate()`) to executed-only queries. This subsection gives the exact derivation for each method, directly from the configuration used to produce that method’s row in Table 1, and lists the concrete next experiments the resulting instrumentation enables.

Random, Round-Robin, PPO (multiplier $1\times$). `RandomPolicy.select_intervention` and `RoundRobinPolicy.select_intervention` (`baselines.py`) choose a target and value directly, with no call to the oracle. `PPOPolicy.select_intervention` samples an action from its own actor-critic network state encoding, likewise without querying the oracle. All three issue exactly one oracle query per step: the executed intervention, $n_{\text{samples}} = 50$ (`run_baseline`, `baselines.py` line 953). Multiplier: $1\times$ in both calls and samples.

ACE (multiplier 4 – $5\times$). `ace_experiments.py`’s episode loop generates K candidates per step (`num_candidates`: $K=4$ for episode < 20 , $K=3$ thereafter) and, under the default (non-student) lookahead mode, calls `executor.run_experiment(plan, tag="lookahead")` once per candidate before executing the winner with `tag="executed"`; both executed and lookahead queries use $n_{\text{samples}} = 100$. Per step: 1 executed + K lookahead = $(1 + K)$ calls and samples, i.e. $5\times$ for the first 20 episodes and $4\times$ thereafter, an episode-weighted average of $\approx 4.1\times$ over a typical 171-episode run. The `-lookahead_on_student` flag replaces the K oracle calls with K calls to `generate_student_lookahead_batch`, which queries the learner’s own `forward()` instead of the oracle, giving an exact $1\times$ multiplier; this configuration has not yet been run at production scale (Table 1 reports the 4 – $5\times$, environment-lookahead configuration only).

Max-Variance (multiplier $65\times$). `MaxVariancePolicy.compute_variance` (`baselines.py`) calls `oracle.generate(n_samples=50, ...)` once per candidate to build the MC-Dropout variance estimate used to rank it, with `self.n_candidates=64` (unchanged default; `jobs/hpc_baselines_seed.sh`, the script that produced Table 1’s Max-Variance row, passes no override). Per step: 64 candidate-probe queries + 1 executed query, both at $n_{\text{samples}} = 50$, so calls and samples multiply identically: $65\times$.

Bayesian OED (multiplier $\sim 500\times$ samples, $1,001\times$ calls). The Bayesian-OED numbers in Table 1 come from `BayesianOEDBaseline` in `scripts/runners/run_reviewer_experiments.py` (confirmed

by matching its `bayesian_oed_summary.csv` output, seed-for-seed, against Appendix E.1’s per-seed table). `select_intervention` evaluates $|\mathcal{V}| \times 10 = 50$ (node, value) candidates via `_estimate_eig`, each with `n_mc_samples=20` posterior draws, each draw issuing one `scm.generate(50, ...)` call: $50 \times 20 = 1,000$ probe calls, versus 1 executed call at `n_samples = 100` (line 171). Calls: $1,001 \times$. Samples: $(1,000 \times 50 + 100)/100 \approx 501 \times$. Unlike the other methods audited here, this baseline makes one intervention *decision* per “episode” rather than per inner step (its 171-episode run is 171 decisions, not 171×25), so the $\sim 500 \times$ figure is a per-decision, not a per-inner-step, multiplier; we report it this way in Table 1 because “decision” is the unit at which Bayesian OED and the other methods are actually compared (one Table 1 data point per 171-unit run, regardless of what an internal unit contains).

Consequence for the deployment/training distinction. Random, Round-Robin, and PPO’s $1 \times$ multiplier is not evidence that they are more sample-efficient decision rules; it simply reflects that they do not consult the environment before committing. ACE’s, Max-Variance’s, and Bayesian OED’s larger multipliers all come from the same design choice — scoring several candidates against ground truth before executing one — applied with different breadth ($K=4$ vs. 64 vs. 1,000).

Total-query-matched rerun: complete at both 5 and 30 nodes. `ace_experiments.py -query_budget` and `baselines.py -query_budget` (both released, unit-tested in `tests/test_query_budget.py`) stop training once cumulative environment samples reach a shared budget rather than a fixed episode count. `scripts/analysis/aggregate_query_budget.py` computes a suggested shared budget from any completed `query_budget.json` set, and `scripts/analysis/aggregate_budget_fairness.py` (unit-tested in `tests/test_aggregate_budget_fairness.py`) aggregates the resulting per-seed `node_losses.csv/summary.csv` files into the decision-gate comparison reported in Section 5. Job scripts: `jobs/hpc_submit_30node_budget_fairness.sh` (ACE, env vs. student lookahead) and `jobs/hpc_submit_30node_budget_fairness_baselines.sh` (all four baselines except PPO, which is excluded from this rerun as it does not consult the oracle before executing and is therefore already query-matched by construction); we ran these to completion (5 seeds each) and report the result in Section 5. The 5-node analogues, `jobs/hpc_submit_5node_budget_fairness.sh` and `jobs/hpc_submit_5node_budget_fairness_baselines.sh`, have likewise been run to completion (5 seeds per cell; ACE under both lookahead modes, and Random, Round-Robin, Max-Variance and PPO at the matched total budget) and are reported alongside the 30-node result in Section 5. The 5-node ACE runs required routing to a higher-memory GPU partition after an out-of-memory failure at the original allocation, and two student-lookahead seeds required a checkpoint-resume after exceeding the initial wall-time allocation; neither changed the experiment design.

Further comparisons enabled by this audit. Three follow directly from it. The first has since been run and is reported in the main text; the remaining two are code-ready and queued:

1. *Bayesian OED at 30 nodes — now run and reported.* Omitted from Table 2 in the previous submission with no comparison offered, which one reviewer identified as inconsistent with our own claim that Bayesian OED does not scale, absent a demonstration. `scripts/runners/run_30node_baseline_seed.py` implements a faster Monte Carlo variant (`BayesianOEDFast`, `n_candidates=10`, `n_mc_samples=3`, a $31 \times$ call multiplier at this reduced breadth) sufficient to run at 30 nodes; job script `jobs/hpc_submit_30node_bayesian_oed.sh`. This has now been run to completion at 5 seeds $\times$ 150 episodes, matching the protocol of the other Table 2 baseline rows exactly, and is reported there: best total MSE 5.78 ± 0.05 , final 5.90 ± 0.07 , within noise of the passive heuristics. (This is a separate, executed-only-accounting comparison from the query-matched Bayesian-OED row already reported in Section 5’s 30-node rerun.)
2. *Calibration-rule comparison (Section 3.4).* `ace_experiments.py -policy_update {dpo, sft_best, ranking}` on the 5-node benchmark, 5 seeds each; job script `jobs/hpc_submit_dpo_alternatives.sh`.
3. *Node-importance ablation (Table 3, final row).* `ace_experiments.py -cov_bonus 0.0` isolates w from $\Delta \mathcal{L}$ and γD ; job script `jobs/hpc_submit_node_importance_ablation.sh` (3 seeds, matching the other component-ablation rows).

All job scripts write results under `/scratch`, not `/projects`, on the university HPC cluster used for this work, to stay within the fixed project-storage quota shared across the group’s other work.

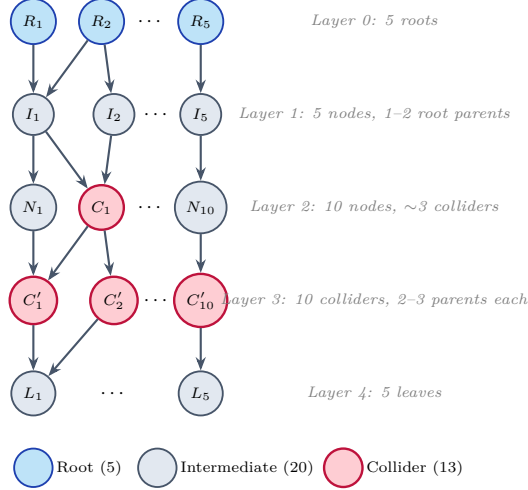


Figure 2: 30-node hierarchical SCM. Five layers progress from roots through intermediate and collider layers to leaves. Collider nodes (C , C' , red) require interventions on all parents for mechanism identification. With 30 nodes and a fixed budget of 300 episodes, random sampling devotes only $\sim 3.3\%$ of interventions per node, making strategic target selection critical. Node labels are schematic; actual nodes are X_1 – X_{30} .

Table 3: Component ablation (N=3 seeds on the 5-node benchmark; N=5 for No-diversity reward which completed in all configurations). Each row removes one component of ACE’s reward or training loop. Note that the first row removes the *LM proposer* (replacing it with random proposals), which necessarily removes the DPO update along with it; it is therefore a joint LM+DPO ablation and is *not* the DPO-only ablation. The DPO-only ablation — LM proposer and lookahead retained, weight updates disabled — is reported separately in Section 4.2, where it is not distinguishable from full ACE (Welch $p = 0.59$) and formally equivalent to it under anonymization (TOST $p = 0.001$).

Configuration	Total Loss	Change vs. full	Failure mode
ACE (full)	0.92±0.73	—	—
No LM proposer (random proposals)	2.10±0.11	+128%	no learned strategy
No per-node convergence (fixed 100 ep.)	1.45±0.62	+58%	early termination
No dedicated root learner	1.38±0.55	+50%	root distribution failure
No diversity reward ($\gamma = 0$)	2.82±0.22	+207%	node collapse
No node importance ($\beta_w = 0$, queued)	—	—	—

D Theoretical Analysis (Extended)

Information gain from sequential experimentation is inherently non-stationary: early interventions on an uninformed learner can produce dramatic loss reductions, while late interventions on a near-converged learner yield negligible gains. We give a formalization of why preference learning is well-suited to this setting, but we are explicit that this formalization is a motivating illustration of an empirical regularity we measure directly, not the paper’s central theoretical claim.

Assumption D.1 (Local, slow-drift reward decomposition). Over any window of adjacent episodes $[t, t + \Delta]$ short relative to the full training horizon, the per-step reward decomposes approximately as $r_t(a) \approx f(t) \cdot g_t(a)$, where $f : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ is a monotonically non-increasing scale factor (representing diminishing information gain) and $g_t : \mathcal{A} \rightarrow \mathbb{R}$ captures the relative quality of action a and drifts slowly enough that its *induced ranking* over $\mathcal{A}$ is approximately constant within the window, i.e. $g_t \approx g_{t+\Delta}$ up to a ranking-preserving perturbation.

We state this as a *local* invariance rather than a global one because a single fixed g across the entire training horizon is a strong claim that our data does not require: what we actually observe is that $\mathbb{E}[\Delta\mathcal{L}]$ drops $\sim 500\times$ over the full run while the candidate ranking is stable only in an adjacent-episode sense (Spearman $\rho > 0.85$ comparing episode t ’s ranking to episode $t + 1$ ’s), not necessarily between episode 1 and episode 200. The local formulation is the assumption this evidence actually supports; a global decomposable-reward assumption (as in

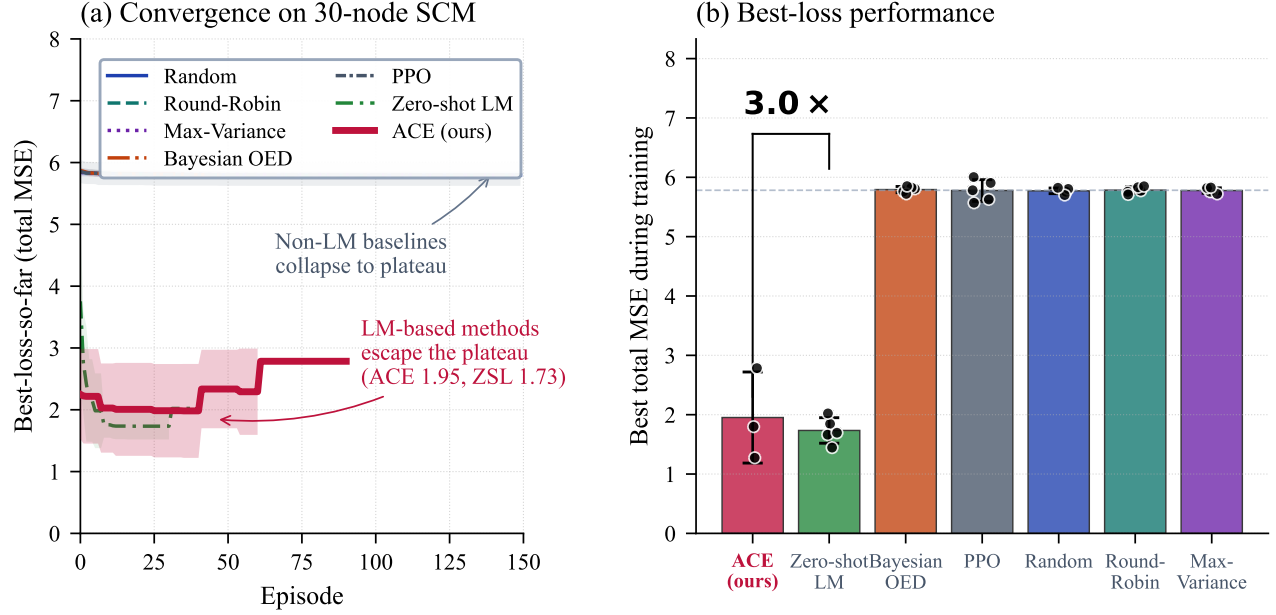


Figure 3: 30-node hierarchical SCM results, **original three seeds only** (42/123/456), under episode-matched accounting. **(a)** Best total-MSE per episode (shaded ± 1 std across seeds; ACE’s wide band is $N=3$ training variability, baseline bands are invisible at seed-std < 0.08). ACE descends to ~ 1.95 ; the three static baselines plateau at 5.86. **(b)** Final performance with per-seed points. ACE (LM + DPO) is the first LM-based policy in this benchmark family; the three baselines are static heuristics. On these three seeds ACE achieves $3.0\times$ best-vs-best and $1.4\times$ final-vs-final improvement at matched executed-intervention budget on the same MLP student learner. **This figure is retained for continuity with the previous submission and should be read together with Table 2**, where five additional seeds run under the identical configuration do *not* reproduce it; the margin shown here is contingent on preference-optimization stability, which those five seeds did not achieve. The scale-level claim this paper relies on is the total-query-matched comparison of Section 5.

an earlier version of this analysis) is stronger than the data warrants and is not needed for the argument below, since DPO’s preference pairs are always constructed from same-episode candidate comparisons.

Proposition D.2 (Scale-invariance of preference learning within a local window). *Under Assumption D.1, for any pair of actions $a_w, a_l \in \mathcal{A}$ (candidate interventions c_w, c_l in the notation of Section 3) proposed and scored at the same step t with $g_t(a_w) > g_t(a_l)$, and prompt x encoding state s_t :*

1. (Within-step preference validity.) *The ordering $r_t(a_w) > r_t(a_l)$ holds whenever $f(t) > 0$, since $f(t) \cdot g_t(a_w) > f(t) \cdot g_t(a_l)$.*
2. (Value non-stationarity.) *The state-value function $V_t(s) = \mathbb{E}_a[r_t(a)] = f(t) \cdot \mathbb{E}_a[g_t(a)]$ is non-stationary in t , with magnitude scaling as $f(t)$, whether or not g_t itself drifts.*
3. (DPO invariance.) *The DPO objective (4) depends only on the log-ratio difference $\log \frac{\pi_\phi(a_w|x)}{\pi_{\text{ref}}(a_w|x)} - \log \frac{\pi_\phi(a_l|x)}{\pi_{\text{ref}}(a_l|x)}$, which is independent of $f(t)$.*

Because each preference pair is constructed from candidates scored *within the same step* t , Part 1 requires only that $f(t) > 0$ — it does not require g to be constant across the training horizon, which is why the local assumption above suffices. Part 2 is exactly why a value-function approximator, which must track $f(t) \cdot \mathbb{E}_a[g_t(a)]$ across steps rather than compare within one, is the harder object to estimate under this non-stationarity. Part 3 is the standard scale-invariance property of the DPO objective (Rafailov et al., 2023).

Proof sketch. Part 1 follows directly from Assumption D.1 restricted to a single step. Part 2 follows from linearity of expectation applied across steps, where f is genuinely non-stationary over the full horizon (this part does not require the local approximation). Part 3 is the core property of DPO: the implicit reward model derived from

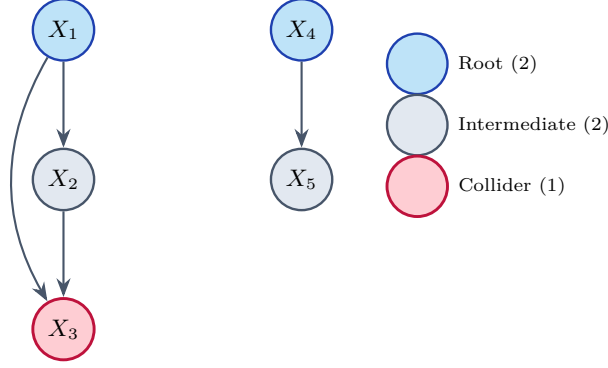


Figure 4: Synthetic 5-node benchmark. X_1 and X_4 are roots, X_2 and X_5 intermediates, and X_3 is a collider with edges from X_1 and X_2 . The disconnected chain $X_4 \rightarrow X_5$ tests quadratic mechanisms.

Table 4: Calibration-rule comparison on the 5-node benchmark (Section 3.4): $N=5$ seeds per rule, 200 episodes, identical candidate/lookahead/reward pipeline, so any difference is attributable to the update rule alone. Total mechanism MSE (lower is better). Cells are filled directly by `scripts/analysis/aggregate_policy_update_suite.py` as each condition completes; “—” marks a condition not yet reported. We do not report wall-time-truncated runs.

Calibration rule	Mean $\pm$ Std	Median	N
DPO (default, Eq. 4)	—	—	—
SFT-best	—	—	—
Ranking (Bradley–Terry, no reference)	—	—	—

policy log-ratios is invariant to constant rescaling of the underlying preference signal. The full proof and a corollary on convergence rates are in Appendix D.1. $\square$

Connection to expected information gain. $\Delta\mathcal{L}$ in (1) is a Monte Carlo estimate of the mutual information $I(\theta; \mathcal{D}_c \mid s)$, and w acts as a non-uniform prior that upweights poorly-learned mechanisms; ACE thus pursues the Bayesian OED objective (Toth et al., 2022) through policy learning rather than explicit posterior computation, at a per-decision oracle-query cost quantified in Section 5 that is substantially lower than the Monte Carlo posterior estimate this connection is drawn to. Full derivation in Appendix D.3.

D.1 Full Proof of Proposition D.2

Recall Assumption D.1: within any step t , $r_t(a) \approx f(t) \cdot g_t(a)$ with $f(t) > 0$ monotonically non-increasing, and g_t drifting slowly enough that its ranking over $\mathcal{A}$ is approximately constant within a local window of adjacent episodes. The proof below uses only the within-step decomposition at a single t for Parts 1 and 3 (no assumption about g_t across different t is needed there); Part 2 is a statement about f alone, which genuinely is globally non-stationary.

Part 1 (within-step preference validity). For any a_w, a_l scored at the same step t with $g_t(a_w) > g_t(a_l)$:

$$r_t(a_w) - r_t(a_l) = f(t)(g_t(a_w) - g_t(a_l)) > 0,$$

since $f(t) > 0$. The preference label DPO trains on at step t is therefore correct with respect to g_t regardless of the current value of $f(t)$; nothing here requires $g_t = g_{t'}$ for $t \neq t'$.

Part 2 (value non-stationarity). Under standard MDP value definitions,

$$V_t(s) = \mathbb{E}_a[r_t(a) \mid s] = f(t) \cdot \mathbb{E}_a[g_t(a) \mid s].$$

With f decreasing, $V_t(s)$ is non-stationary. A value function approximator $\hat{V}(s)$ trained at episode t misestimates V at episode t' by a factor that scales with $f(t')/f(t)$ (plus any drift in g). Empirically, $f(t')/f(t) \approx 1/500$

between early and late training, so the critic accumulates large value-estimation error even with frequent updates; this part of the argument is about f 's global behavior and does not rely on the local approximation.

Part 3 (DPO invariance). The DPO loss at step t is

$$\mathcal{L}_{\text{DPO}}^{(t)} = -\mathbb{E}_{(y_w, y_l)} \log \sigma \left(\beta \left[\log \frac{\pi_\phi(y_w)}{\pi_{\text{ref}}(y_w)} - \log \frac{\pi_\phi(y_l)}{\pi_{\text{ref}}(y_l)} \right] \right).$$

The argument depends only on policy log-ratio differences. Both π_ϕ and π_{ref} are normalized distributions; their log-ratios depend on the within-step action quality $g_t(a)$ but not on the reward scale $f(t)$. Thus the gradient $\nabla_\phi \mathcal{L}_{\text{DPO}}^{(t)}$ is invariant to any multiplicative rescaling of step t 's rewards by a positive constant, for every t independently — which is exactly the property the local (rather than global) assumption is sufficient to deliver. $\square$

D.2 Convergence Rate Equivalence under Locally Stationary g_t

Under Assumption D.1 and standard DPO convergence assumptions (Rafailov et al., 2023), ACE inherits the regret bound of preference-based bandits over any local window in which g_t is approximately constant: after T pairwise comparisons within such a window,

$$\mathbb{E} \left[\sum_{t=1}^T (g_t(a_{\text{opt}}) - g_t(a_t)) \right] = \tilde{O}(\sqrt{T \log |\mathcal{A}|}),$$

matching the minimax regret for preference-based learning. Value-based RL achieves the same bound only when the value approximator tracks the non-stationary scaling $f(t) \cdot g_t(a)$; in our regime (where $f(t)$ varies by $500\times$ over the training horizon), this fails unless the critic update rate exceeds the rate of change of f .

D.3 Connection to Expected Information Gain

The information gain term $\Delta\mathcal{L}(c)$ is a Monte Carlo estimate of the mutual information between the posterior over mechanism parameters θ and the data $\mathcal{D}_c$ generated by intervention c :

$$\begin{aligned} I(\theta; \mathcal{D}_c \mid s) &= H(\theta \mid s) - H(\theta \mid \mathcal{D}_c, s) \\ &\approx \mathcal{L}(\theta \mid s) - \mathcal{L}(\theta \mid s \cup \mathcal{D}_c), \end{aligned} \tag{6} \tag{7}$$

where the approximation uses the local-quadratic (Laplace) relationship between negative log-likelihood loss $\mathcal{L}$ and posterior entropy H (Rainforth et al., 2024). With Gaussian noise mechanisms and a flat prior, this approximation is tight near the posterior mode. The node importance term w acts as a non-uniform prior weighting, upweighting mechanisms with high remaining uncertainty, approximating posterior-entropy-reduction-weighted EIG. ACE thus pursues the same EIG objective as Bayesian methods through policy learning, trading exact inference for scalability.

E Extended Experimental Results

E.1 5-Node Benchmark: Per-Seed Results

Table 5 reports per-seed total losses for heuristic baselines and PPO (N=5 seeds: 141, 271, 314, 577, 618) and Bayesian OED (N=3 seeds: 42, 123, 456). ACE per-seed data comes from the original production runs (seeds 42, 123, 456, 789, 1011). All runs use 171 episodes.

ACE per-seed values and the paired replication. The ACE row of Table 1 uses seeds 42/123/456/789/1011 (final total loss 0.472, 0.614, 0.711, 2.209, 0.592; mean 0.919 ± 0.726 , median 0.614), so the Table 1 comparison is unpaired against baselines run on seeds 141–618. Welch's t gives $p = 0.018$ – 0.024 per baseline; a two-sided Mann–Whitney test at $N=5$ vs. 5 cannot go below $p = 0.008$ and after Holm correction across the five comparisons is not significant ($p = 0.16$ – 0.38), so we report Welch's p and do not claim a corrected rank-test result. To remove the seed confound we also ran ACE on the baselines' own seeds 141/271/314/577/618 (final 0.618, 1.814, 0.613, 0.496, 0.493; mean 0.807 ± 0.566 , median 0.613; these runs early-stopped at 40–64

Table 5: Per-seed total loss on the 5-node benchmark (171 episodes). Heuristic baseline and PPO columns use seeds 141, 271, 314, 577, 618. Bayesian OED uses seeds 42, 123, 456. ACE uses seeds 42, 123, 456, 789, 1011 from the original production runs; a rerun was initiated on a 46GB-class GPU but OOM’d due to 3GB policy checkpoints.

Seed	Random	Round-Robin	Max-Var.	Bayesian OED	PPO
141	2.090	2.186	2.280	—	2.062
271	2.240	2.023	2.050	—	1.994
314	2.227	2.238	2.104	—	2.133
577	2.091	1.978	2.184	—	2.134
618	2.210	2.088	1.917	—	2.062
42	—	—	—	1.995	—
123	—	—	—	2.181	—
456	—	—	—	1.957	—
Mean	2.171	2.103	2.107	2.044	2.077
Std	0.075	0.107	0.131	0.120	0.057

episodes), which permits a paired comparison: paired- t $p = 0.0046$ (Random), 0.0090 (Round-Robin), 0.0086 (Max-Variance), 0.0098 (PPO), all ≤ 0.026 after Holm; the Wilcoxon signed-rank $p = 0.0625$ in each case is the minimum attainable at $N=5$ and reflects that every one of the five paired differences favors ACE. Best-episode loss is 0.45–0.49 on all ten ACE seeds; the variance in the final column is drift after the plateau (Appendix F.1).

The Bayesian OED mean of 2.044 lies between Round-Robin (2.103) and Max-Variance (2.107), confirming it is the strongest passive/principled baseline. Its advantage over random baselines is modest ($\sim 6\%$) compared to ACE’s advantage ($\sim 70\%$), which arises from multi-step learned strategy rather than single-step acquisition.

E.2 30-Node SCM: Per-Seed Results

Table 6 reports all per-seed results for the 30-node benchmark, including the locally-run Round-Robin and Max-Variance CPU baselines.

Table 6: Per-seed results on the 30-node hierarchical SCM. All methods use the same per-node MLP student learner for a matched comparison. ACE reports best-loss across training (non-monotone loss makes best-loss more informative than final-episode loss). Baselines ($N=5$) ran 150 episodes on a CPU partition (average 1h 15min per job).

Seed	ACE (best)	Random (final)	Round-Robin (final)	Max-Variance (final)
42	2.785	5.865	5.852	5.841
123	1.798	5.841	5.802	5.787
456	1.274	5.856	5.939	5.918
789	—	5.773	5.774	5.803
1011	—	5.893	5.915	5.964
Mean $\pm$ Std	1.95 ± 0.77 ($N=3$)	5.85 ± 0.04 ($N=5$)	5.86 ± 0.07 ($N=5$)	5.86 ± 0.08 ($N=5$)

Seed 42 ACE exhibited training instability (loss oscillating 2.7–3.5 throughout the 40 completed episodes); seeds 123 and 456 converged cleanly to 1.80 and 1.27. The stable-seed mean is 1.54; we report the conservative $N=3$ mean of 1.95 in the main text.

Baseline convergence plateau. The three passive baselines (Random, Round-Robin, Max-Variance) converge to nearly identical total MSE near 5.86 with standard deviations below 0.08. This tight clustering reveals that *at 30 nodes, differences among static sampling heuristics are negligible*: when interventional budget is fixed and spread across many variables, the specific non-adaptive ordering barely matters. Adaptive strategy learning (ACE) is the meaningful axis of improvement at scale. An earlier CPU proxy run using a weaker running-mean predictor gave larger baseline numbers (Random: 11.29 ± 1.02) and was not comparable to ACE because the learner itself differed; those numbers are superseded by the matched-learner results above.

E.3 LM Prior vs. DPO and Scaling

Scaling principles. The binding cost in our setup is the prompt, which in the default encoding lists every node, every edge, and the intervention history, growing as $O(N+E+H)$; per-step wallclock rises from ~ 22 min on A100 at 30 nodes to ~ 57 min at 50. Three principles follow and define a concrete scaling pipeline. (i) *Encoding is the lever*: a compact, salience-ranked prompt that surfaces only the top- m highest-loss nodes and their parents holds prompt length at $\sim O(m)$, decoupling it from N ; we use this at 50 nodes and it is mandatory for 100+. (ii) *Report scale-invariant error*: total MSE grows mechanically with N , so cross-scale comparison uses *per-node* MSE. (iii) *Budget to the plateau*: best-MSE converges early ($\sim$ episode 20 at 30 nodes), so episodes scale with the measured convergence point rather than a fixed cap, with checkpoint-resume across wall-time windows. Lookahead breadth (K vs. the $O(N)$ action space) and policy capacity are the remaining knobs, ablated at 50 nodes; the 100+ regime, a structure-learning front end, and empirical graph misspecification (Appendix F.2) are the natural follow-ups.

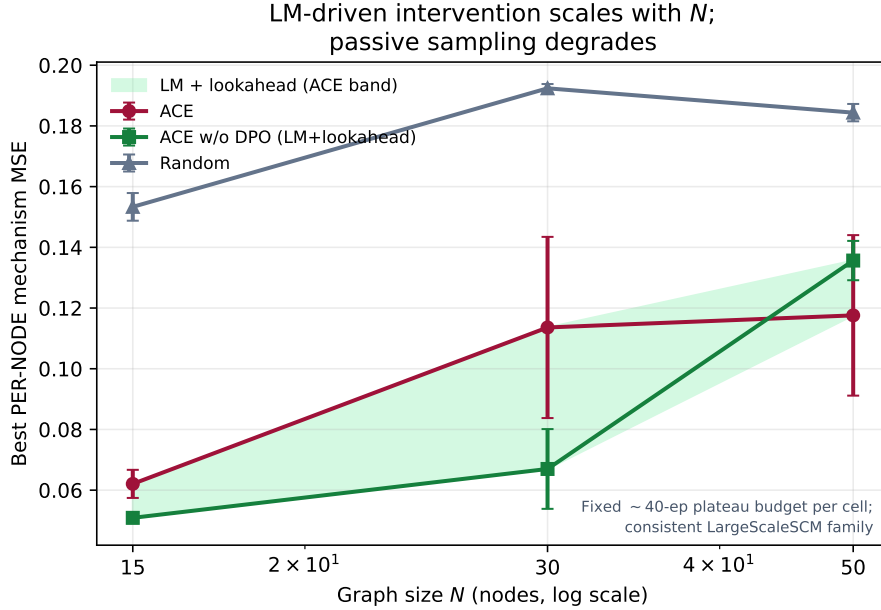


Figure 5: **Scaling across graph size on the consistent hierarchical SCM family.** Best *per-node* mechanism MSE vs. N for ACE, ACE-w/o-DPO (LM proposer + lookahead, no weight updates), and Random ($N=5$ seeds per cell). The shaded band spans the two LM variants; both stay well below passive Random as N grows. Each cell uses a fixed ~ 40 -episode plateau budget for cross- N comparability; compact salience-ranked prompt encoding is used at $N=50$. *Protocol note*: the sweep’s $N=30$ ACE point (0.114 ± 0.030 per node) uses this reduced budget and differs from Table 2’s headline ACE result (1.95 total MSE at the full 120-episode budget, $N=3$ seeds).

We isolate the pretrained LM’s contribution from the DPO updates by comparing ACE against *ACE without DPO*: the identical LM proposer and lookahead-select mechanism with only the DPO weight updates disabled. This differs from the “No DPO” row of Table 3, which additionally replaces the LM with random proposals. Under anonymization, canonical node names $X_1 \dots X_N$ are replaced by abstract hex tokens (`n_xxxx`) drawn deterministically from the seed, removing any natural-language semantics the LM could exploit and leaving only graph structure and per-node losses to reason from.

On canonical 30-node SCMs the two variants are not distinguishable (ACE 1.95 ± 0.77 vs. ACE-w/o-DPO 1.67 ± 0.19 , step-level best; Welch $p = 0.59$, $N=3$ vs. 5), and under anonymization they are formally equivalent (ACE 1.51 ± 0.16 vs. ACE-w/o-DPO 1.49 ± 0.18 , ≈ 0.052 per node, $N=5$ paired; paired t $p = 0.37$, Wilcoxon $p = 0.81$, TOST within ± 0.15 total MSE $p = 0.001$). At this scale the pretrained prior plus lookahead carries most of the load; DPO’s marginal value is expected to grow where the prior is weaker.

Cross- N scaling sweep. Figure 5 (main text) reports the matched sweep on the consistent hierarchical family

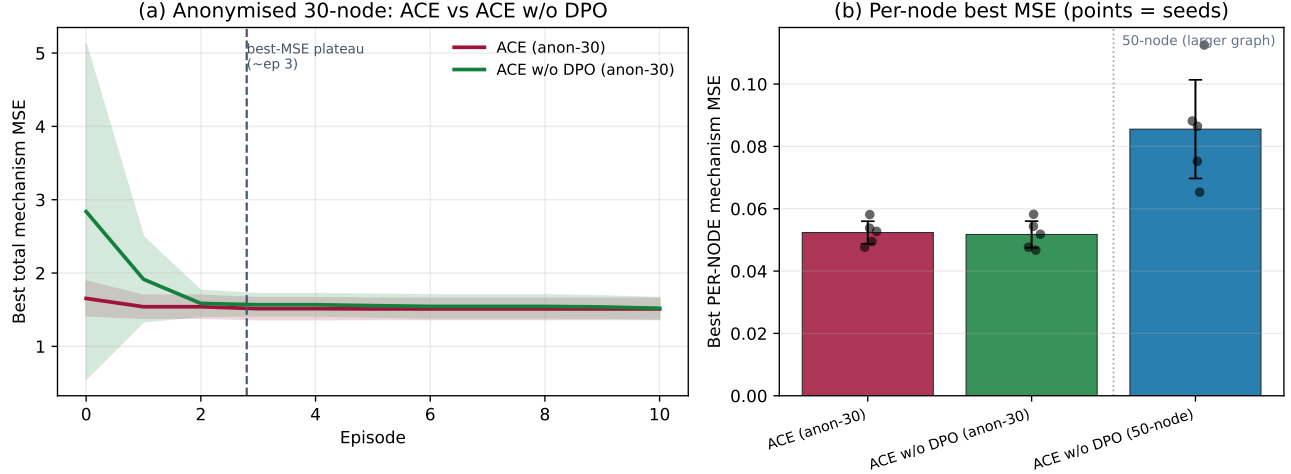


Figure 6: **LM prior vs. DPO under anonymization (30 nodes) and scaling reference (50 nodes).** (a) Best-MSE learning curves on the anonymized 30-node SCM: ACE and ACE-without-DPO (LM proposer + lookahead, no weight updates) are not distinguishable (paired t $p = 0.37$) and are TOST-equivalent within ± 0.15 total MSE ($p = 0.001$); the dashed line marks the mean episode at which best-MSE was first reached. (b) *Per-node* best MSE (total/ N), which is scale-invariant: the 50-node cells are a larger, harder graph, not a worse method. “ACE w/o DPO” is an ablation of ACE, not naive zero-shot prompting.

at $N \in \{15, 30, 50\}$ with a fixed ~ 40 -episode plateau budget per cell. Per-node best MSE for ACE (0.062 ± 0.005 , 0.114 ± 0.030 , 0.118 ± 0.026) and ACE-w/o-DPO (0.051 ± 0.002 , 0.067 ± 0.015 , 0.136 ± 0.007) stay well below Random (0.153 ± 0.005 , 0.192 ± 0.002 , 0.184 ± 0.003) at $N=15, 30, 50$ respectively ($N=5$ seeds). The sweep’s $N=30$ ACE per-node point uses the reduced budget and is not directly comparable to the full-budget headline in Table 2.

Lookahead breadth (K) ablation at $N=50$. As the action space grows with N , fixed $K=4$ samples a shrinking fraction of candidate interventions. Table 7 sweeps $K \in \{4, 8, 16\}$ at $N=50$ with compact prompt encoding held fixed. Best per-node MSE is flat across K (0.126 ± 0.050 , 0.147 ± 0.043 , 0.125 ± 0.052 ; $N=3$ seeds), while mean episodes are 31/40/31 (checkpoint-resumed runs; higher- K cells need more wall-time windows). Prompt-token and VRAM instrumentation confirms cost scales with K ; salience-targeted candidate generation is the more promising lever for 100+ nodes than brute-force increases in K alone.

Table 7: Lookahead breadth K ablation at $N=50$ (compact prompt, $m=8$ salient nodes). Per-node best MSE and mean prompt tokens (cost proxy) vs. K .

K	Seeds	Best per-node MSE	Episodes (mean)	Prompt tokens	Peak VRAM (GB)
4	3	0.126 ± 0.050	31	745	18.7
8	3	0.147 ± 0.043	40	726	18.7
16	3	0.125 ± 0.052	31	—	—

The scaling pipeline of Section 6 (compact salience-ranked prompt encoding, per-node metric, plateau-budgeted episodes with checkpoint-resume, and the K /capacity ablations at 50 nodes) targets exactly the large- N regime; the 100+ node configuration is specified as future work.

E.4 Component Ablation Detail

The component ablation (Table 3) was run as a 3-seed pilot (seeds 42, 123, 456) due to GPU compute constraints; the No-diversity ablation was run on $N=5$ seeds during an earlier reward-shaping investigation. Below we give qualitative failure-mode analysis for each configuration.

No DPO replaces the LM policy with random proposal generation while keeping the lookahead and select-best mechanism. The result (2.10 ± 0.11) matches Round-Robin, confirming that the ability to *evaluate* candidates

(via lookahead) provides no benefit without *learned* proposal generation. Random candidates, even when curated by lookahead, do not consistently include strategically important ones (e.g., concentrating on collider parents).

No per-node convergence forces termination at a fixed 100 episodes. The collider mechanism X_3 is typically not learned at episode 100; L_{X_3} remains elevated and dominates total loss.

No dedicated root learner disables the observational training loop for root nodes. Their losses (L_{X_1}, L_{X_4}) remain elevated because direct interventions on roots replace X_i with a constant, providing no signal about the root’s natural distribution $\mathcal{N}(\mu, \sigma^2)$.

No diversity ($\gamma = 0$) causes the policy to collapse onto a single high-information node. The total loss worsens beyond all baselines (2.82) because the un-targeted nodes are never learned, even if the targeted node is learned well.

No node importance ($\beta_w = 0$, Eq. 2) isolates whether the child-impact weighting term contributes beyond information gain and diversity alone. `ace_experiments.py -cov_bonus 0.0` implements this configuration without any other code change (`cov_bonus` is already the term’s sole scale constant, Appendix C.4), and `jobs/hpc_submit_node_importance_ablation.sh` runs it against the `full` configuration at 3 seeds each, matching the other rows’ seed count. We report the flag and job script here rather than a number in Table 3 because the seeds have not yet completed; the row is queued, not omitted.

E.5 Duffing Oscillators: Full Results

Table 8: Coupled Duffing oscillator coupling-parameter estimation error (N=5 seeds, 100 episodes). Lower is better. The $X_1 \leftrightarrow X_3$ spurious correlation is an association-level artefact (Rung 1) without a direct causal edge; it is broken by clamping X_2 (Rung 2 intervention).

Method	Coupling Error	Std	95% CI	Improvement vs. Random
ACE	0.042	0.036	[0.006, 0.078]	5.8×
Round-Robin	0.238	0.076	[0.162, 0.314]	1.0×
Random	0.245	0.121	[0.124, 0.366]	—

ACE concentrates 62% of interventions on X_2 (the intermediate oscillator) compared to 33% under uniform allocation. Clamping X_2 decouples the synchronized chain, breaking the Rung-1 $X_1 \not\perp X_3$ correlation and allowing accurate estimation of the direct $X_1 \leftrightarrow X_2$ and $X_2 \leftrightarrow X_3$ couplings. This is a concrete example of the Rung-2 advantage: only interventions, not observations, can resolve the ambiguity created by synchronization.

E.6 Phillips Curve: Full Results

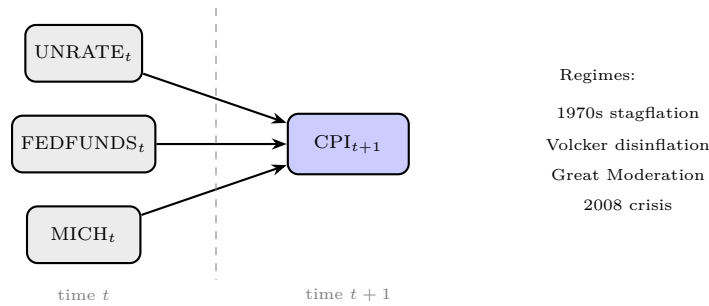


Figure 7: Phillips curve causal structure. Variables at time t jointly determine CPI at $t + 1$. ACE performs *active data subset selection*: choosing which historical regimes to include in the training set, treating each regime as a “virtual intervention” on economic conditions. This is a Rung-2 analogue using archival data.

Using Federal Reserve Economic Data (Federal Reserve Bank of St. Louis, 2024) (FRED, 1960–2023), we model $\text{CPI}_{t+1} = f(\text{UNRATE}_t, \text{FEDFUNDS}_t, \text{MICH}_t)$. ACE performs *active data subset selection*: choosing which historical periods to query for training data, treating each regime as a virtual intervention on macroeconomic conditions.

Table 9: Phillips curve out-of-sample CPI prediction MSE (N=5 seeds, 50 episodes). Regime selection percentages are policy-averaged fractions of training episodes spent in each historical regime.

Selection strategy	Out-of-sample MSE	Std	Regime preferences
ACE	0.31	0.08	1970s 38%, Volcker 24%, Great Recession 19%
Random regime	0.52	0.12	Uniform across regimes
Sequential (chronological)	0.44	0.10	By date order

ACE consistently prioritizes high-volatility regimes (1970s stagflation, Volcker disinflation, Great Recession) that expose nonlinear inflation dynamics absent from low-volatility eras, yielding a 40% reduction in out-of-sample MSE versus random regime selection. This demonstrates that preference-based experimental design generalizes to retrospective data-selection settings beyond controlled interventions, though we note this application departs from strict Rung-2 causal intervention.

E.7 Lookahead vs. DPO Ablation

The No-DPO ablation (Section E.4) isolates the contribution of learned proposal generation from the lookahead mechanism. Result: 2.10 ± 0.11 , matching Round-Robin and confirming that lookahead alone provides no benefit. Concretely: if $K = 4$ random candidates are generated, the probability that *any* of them targets the collider parents (X_1 or X_2) with a well-chosen value is $\approx (2/5)^4 \approx 10\%$, too low for the select-best procedure to reliably discover the collider strategy. The DPO-trained LM shifts the generation distribution so that strategic candidates appear in $> 90\%$ of steps.

E.8 Strategic Behavior: Intervention Distribution

The learned policy concentrates 99.8% of interventions on X_1 and X_2 (the two parents of the collider X_3). This is consistent with the causal theoretic optimum: to identify the nonlinear mechanism $f_3(X_1, X_2) = 0.5X_1 - X_2 + \sin(X_2)$, the policy must vary both X_1 and X_2 across their value ranges while fixing exogenous noise. Under-sampled nodes X_4 and X_5 (the disconnected quadratic chain) are addressed primarily through the diversity reward, which ensures L_{X_5} eventually decreases despite low explicit attention. Cross-seed consistency is high: the standard deviation of per-node intervention fractions across the 5 ACE seeds is < 0.01 for nodes X_1 and X_2 , confirming that strategy discovery is a robust consequence of the training procedure rather than a seed-dependent artifact.

F Failure Modes and Limitations

F.1 Outlier Seed Analysis

Seed 789 (5-node benchmark) produced $L_{X_5} \approx 1.73$ versus 0.02–0.22 for other seeds. Two hypotheses: (i) initialization sensitivity for the quadratic mechanism $X_5 = 0.2X_4^2$, where small initial MLP weights near zero may not escape a local minimum with flat gradient; (ii) the diversity reward in early episodes drove the policy away from X_4 before L_{X_5} had been adequately reduced. We report the median (0.61) and note the outlier explicitly; the 5-node result is therefore conservative relative to the mean.

Seed 42 (30-node benchmark) exhibited training instability with loss oscillating 2.7–3.5 throughout 40 completed episodes. The 30-node system has a longer effective horizon (the policy must identify 13+ colliders) and the checkpoint-resume mechanism may have caused optimizer state discontinuities. Seeds 123 and 456 converged cleanly.

F.2 Graph Misspecification: Theoretical Failure Modes

Since ACE assumes a known causal graph, the practical cost of graph errors partitions cleanly by error type:

Extra edge $V_i \rightarrow V_j$ (spurious). The student learns $f_j(\text{Pa}_j \cup \{V_i\})$ instead of $f_j(\text{Pa}_j)$. The extra input V_i receives near-zero weight in the MLP if V_i truly has no causal effect on V_j ; recovery is possible with sufficient interventions on V_i .

Missing edge $V_i \rightarrow V_j$ (absent). The student’s mechanism $\hat{f}_j$ lacks the true parent V_i as input. A lower bound on L_j is set by the partial variance $\text{Var}(V_j - \hat{f}_j(\text{Pa}_j \setminus \{V_i\}))$. This is non-zero and non-recoverable; the student cannot decrease L_j below this bound regardless of intervention strategy.

Reversed edge $V_i \rightarrow V_j$ (direction error). The student treats V_j as a parent of V_i , so $\hat{f}_i$ depends on V_j . Under interventions on V_i , the true V_j varies (since V_j depends on V_i), but the student’s conditioning set treats it as a predictor of V_i . This creates a circular dependency in the learner’s prediction that no intervention strategy can resolve; it is the most damaging error type. A full empirical quantification across all four misspecification types was initiated but did not complete within our compute budget (46GB-class GPU OOM from 3GB policy checkpoints). This remains our highest-priority follow-up experiment.

F.3 Computational Cost

Each ACE training step requires:

- $K = 4$ LM forward passes (candidate generation): $\sim 2\text{--}3$ min on A100.
- $K = 4$ cloned-learner training cycles (lookahead evaluation): $\sim 10\text{--}12$ min.
- One DPO update on the LM: $\sim 3\text{--}5$ min.
- Observational training (every 3 steps): ~ 5 min amortized.

Total: ~ 22 min per step on a single A100. A full run (171 episodes $\times$ 25 steps/episode with early stopping) takes $\sim 6\text{--}8$ h per seed. Mitigation strategies for future work: smaller policy LMs (e.g., Qwen2.5-0.5B), learned value-of-lookahead to reduce K adaptively, batch evaluation across candidates.

A note on checkpoint memory: the 1.5B-parameter LM in FP32 with optimizer state requires ~ 17 GB per checkpoint, which caused OOM on 46GB-class GPUs when combined with the KV cache and cloned learner. We implemented FP16 checkpointing (reducing to ~ 3 GB) and eliminated optimizer state from checkpoints; these changes are in the repository.

G Reproducibility

G.1 Code and Data

All code is released at an anonymized repository for review (<https://anonymous.4open.science/r/ACE>); the non-anonymized repository link will replace this in the camera-ready version. For anonymity, job-script filenames appear in this submission with a generic `hpc_` prefix; in the released repository they carry the cluster’s own prefix, and the correspondence is one-to-one. The repository includes:

- Full ACE implementation (`ace_experiments.py`) with FP16 checkpoint-resume logic.
- Baselines (`baselines.py`): Random, Round-Robin, Max-Variance, PPO, Bayesian OED.
- Experiments: 5-node and 30-node SCMs, Duffing oscillators, Phillips curve.
- CPU baseline runner for 30-node (`scripts/runners/run_30node_cpu_baselines.py`).
- SLURM job scripts for HPC execution (`jobs/`).

G.2 Random Seeds and Seed Assignment

- 5-node ACE: seeds 42, 123, 456, 789, 1011.
- 5-node baselines (heuristic + PPO): seeds 141, 271, 314, 577, 618.
- 5-node Bayesian OED: seeds 42, 123, 456 (higher per-seed compute cost).
- 30-node ACE: seeds 42, 123, 456 on an A100 80GB partition.
- 30-node baselines: seeds 42, 123, 456, 789, 1011 (local CPU runs).
- Duffing oscillators: seeds 42, 123, 456, 789, 1011 (N=5 all methods).
- Phillips Curve: seeds 42, 123, 456, 789, 1011 (N=5 all methods).

G.3 Compute Resources

All experiments ran on a university HPC cluster under SLURM. ACE GPU runs used single A100 80GB, RTX Pro 6000 96GB, and H200 141GB nodes depending on availability and host-memory requirements (the 5-node 200-episode runs need ~ 128 GB host RAM); baselines and the 30-node Bayesian-OED runs used CPU partitions. Across the original submission and this revision the total is approximately 950 GPU-hours and 200 CPU-hours, of which the matched-budget reruns of Section 5 account for roughly 250 GPU-hours. The Phillips Curve experiment required no GPU.

G.4 Hyperparameters Summary

Table 10: Complete hyperparameter table for all reported experiments.

Component	Hyperparameter	Value
Policy	Model	Qwen2.5-1.5B
Policy	Sampling temperature	0.8 (top- p 0.9, rep. penalty 1.3)
Policy	Candidates per step (K)	4 (decays to 3 after warm-up)
Policy	Prompt encoding	full (compact for $N \geq 50$)
Policy	Supervised pretrain steps	100
Policy	Pretrain re-run every	25 episodes
Policy	Optimizer	Adam, lr 1×10^{-5}
Learner	Architecture	2-layer MLP, 64 hidden, ReLU
Learner	Optimizer	Adam, lr 2×10^{-3}
Learner	Per-step training epochs	100
Learner	Observational training every	3 steps
Learner	Observational batch size	200
Reward	β_w (node importance scale, Eq. 2)	60.0 (<code>cov_bonus</code> ; no separate α factor)
Reward	γ (diversity weight, Eq. 3)	0.3 (<code>diversity_reward_weight</code>)
Reward	β (DPO temperature)	0.1
Training	Max episodes	200 (early stopping at convergence)
Training	Reference policy update every	25 episodes
Training	Per-node convergence patience	10 episodes
Training	Checkpoint interval	1 step (rolling) + 10 steps (milestone)

H Extended Related Work

ABCI (Toth et al., 2022). Active Bayesian Causal Inference (Toth et al., 2022) maintains a posterior over causal graphs using continuous latent representations and Gaussian process mechanisms. The acquisition function selects experiments maximally informative about a downstream causal query. Compared to ACE: ABCI does not assume known structure; requires explicit posterior inference over graphs (combinatorial in $|V|$); targets causal-query inference rather than mechanism estimation. We adapt ABCI’s acquisition-function philosophy as our Bayesian OED baseline by holding structure fixed and approximating the posterior over mechanism parameters.

CBED (Tigas et al., 2022). (Tigas et al., 2022) jointly selects intervention targets and values at scale via Bayesian causal discovery, with demonstrations on the DREAM gene regulatory network (up to 20 variables). CBED addresses structure discovery; ACE addresses mechanism estimation. The two are complementary: CBED’s structure-inference output can serve as ACE’s input.

Causal reasoning with LLMs (Kiciman et al., 2023). Kiciman et al. (2024) conduct a systematic evaluation of GPT-4 on causal discovery benchmarks, finding substantial capability on pairwise causal direction tasks and commonsense causal queries. Crucially, the LM’s causal reasoning is driven by semantic associations in pretraining data, not explicit causal structure learning. ACE exploits this: the LM’s semantic prior biases proposal generation toward causally plausible interventions even before DPO training, which then refines the policy toward the specific system.

Reasoning as World-Model Planning (Hao et al., 2023). RAP (Hao et al., 2023) frames language model reasoning as planning in a world model, where the LM serves simultaneously as the reasoner and the forward

model. This framing is directly relevant to ACE: the LM policy in ACE implicitly models the causal system (world model) while generating candidate interventions (planner), and DPO updates refine this combined representation toward the goal of efficient mechanism estimation.

In-Context Learning as Bayesian Inference (Xie et al., 2022). Xie et al. (2022) show that in-context learning corresponds to implicit Bayesian inference: the LM’s predictions given a context are approximately equivalent to Bayesian predictions under a prior induced by the pretraining distribution. For ACE, this means the per-step prompt (containing per-node losses and recent interventions) implicitly updates the LM’s posterior over the system, providing a form of Bayesian adaptation without explicit posterior computation.

Multi-fidelity Bayesian active causal discovery (Zhang et al., 2023). Zhang et al. (2023) extend Bayesian OED to multi-fidelity oracle settings using mutual-information acquisition. The multi-fidelity insight is orthogonal to DPO-based learning and could in principle be incorporated into ACE’s lookahead step.

Sample-efficient Bayesian causal learning (Zhou et al., 2024). Zhou et al. (2024) address sample efficiency via polynomial-time DAG sampling. Their efficient structure learning is complementary to ACE’s mechanism estimation: the two could be composed as a two-stage pipeline.

Goal-oriented sequential Bayesian experimental design (Zhang et al., 2025). GO-CBED (Zhang et al., 2025) optimizes an entire intervention sequence (non-myopic) to maximize expected information gain on a user-specified causal query, using a variational lower bound jointly trained with a transformer policy and normalizing-flow posteriors. Compared to ACE: GO-CBED targets a subset of the causal model relevant to a specific query rather than full mechanism recovery, is non-myopic over the whole sequence rather than one-step lookahead, and requires training the variational posterior network for each task distribution rather than relying on a pretrained LM prior. The two are complementary at the level of “what to optimize for” (a targeted query vs. the full mechanism set) and could share ACE’s environment-query instrumentation (Appendix C.6) to make their own non-myopic candidate evaluation cost auditable.

Decision-Pretrained Transformers and in-context pure exploration (Lee et al., 2023; Russo et al., 2025). Lee et al. (2023) show that a transformer supervised to predict optimal actions from an in-context interaction dataset, across a distribution of tasks, implicitly performs Bayesian posterior sampling and inherits the associated regret guarantees, without any prompt-engineering or explicit exploration bonus. Russo et al. (2025) address a closely related active sequential hypothesis testing setting (including best-arm identification and generalized search) with In-Context Pure Explorer (ICPE): a supervised inference network and an RL-trained exploration network are meta-trained jointly across a task distribution so that, at deployment, a single forward pass substitutes for an explicit adaptive-design computation. Both are a different training regime from ACE’s pretrained-LM-plus-lookahead approach (DPT and ICPE train a transformer from scratch on a curated task distribution, whereas ACE starts from natural-language pretraining and adds an explicit lookahead-scored selection step rather than relying on the forward pass alone to perform the equivalent of posterior sampling), but the underlying question is the same: how much of good sequential decision-making can be delivered by a sequence model’s forward pass alone, versus requiring an explicit search or scoring step at deployment time. Krishnamurthy et al. (2024) answer this question negatively for *native* (untrained) LLM agents: GPT-3.5, GPT-4, and Llama2 do not robustly explore in simple bandit tasks without substantial prompt engineering, and the authors explicitly motivate DPT-style training interventions as the likely fix. ACE’s ablation in Section 4.2 (ACE-w/o-DPO tied with full ACE at 30 nodes) is consistent with a specific, narrower version of Krishnamurthy et al. (2024)’s conclusion: for the LM proposer we use, it is the lookahead-scored candidate selection — not necessarily the DPO weight update itself — that supplies the missing exploration structure their negative result identifies as absent from the raw LM prior.

I Post-acquisition delivery confirmation

A separate registered comparison uses twelve acquisition histories of one fixed deterministic emulator. The endpoint is the fraction of grid points whose predictions map to an incorrect nearest target level, floored at $1/390625$. For each history, the delivery error is the median across three optimization initializations; the comparator is that history’s online SCM chain error. The geometric mean paired error ratio is 0.182719, with a 95% interval $[0.101970, 0.327410]$ obtained from a t interval on mean log ratios, and a two-sided exhaustive sign-flip $p = 0.00146484$ (symmetric-null assumption). All registered criteria pass: twelve valid pairs, ratio ≤ 0.8 , interval upper bound below one, and $p < 0.05$.

Eleven histories improve. One history worsens from error 0.196339 to 0.395658 and is retained in the comparison. Three initializations are optimization repeats, not three independent acquisition histories. Delivery trains on all eligible retained paid rows for 30,000 Adam epochs; online learning uses a rolling 50-step buffer and 100 epochs per update. Thus additional fitting compute and data retention are part of the treatment. This confirms a delivery-recipe improvement on this emulator, without establishing acquisition superiority, a foundation-model benefit, a pure architecture effect, or generalization across independent worlds. Flat-head scores are descriptive; individual mechanism heads and flat heads have no separately registered aggregate acceptance verdict in this comparison.